

A novel design of the metal hydride reactor integrated with phase change material for H₂ storage

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Abstract: Using metal hydride for hydrogen storage and transportation is a promising technology due to its advantages of large hydrogen storage capacity, low pressure and low energy consumption. It is common to combine the metal hydride reactor with phase change materials to recover the heat generated during the hydrogen absorption and use it for hydrogen desorption, which can improve the overall energy efficiency. For the reactor design, the heat transfer rate is a key factor affecting the performance of hydrogen absorption and desorption. To enhance heat transfer, this paper proposes a honeycomb fin metal hydride reactor integrated with PCM. Based on simulations, results show that the achieved hydrogen storage capacity is 1.326 wt.%, the gravimetric and volumetric storage densities are 0.411% and 14.76 kg of H₂ per m³, and the mean saturated rates are $1.222 \times 10^{-3} \text{ g s}^{-1}$ and $0.773 \times 10^{-3} \text{ g s}^{-1}$ for absorption and desorption process. In addition, the effects of operational parameters such as fin size, thickness of PCM, hydrogen absorption pressure, and physical properties including solid (liquid) thermal conductivity of PCM on the hydrogen absorption and desorption are investigated. It is found that the small fin size is beneficial to the reaction rate. With the increase of PCM thickness, the mean saturated rates increase, while the gravimetric and

volumetric storage densities decrease. The impact of operating parameters is also significant, increasing the PCM thermal conductivity and changing the initial temperature and hydrogen pressure all can accelerate the reaction rate.

Keywords : Metal hydride ; Hydrogen storage ; Honeycomb fin ; Phase change material; Hydrogen absorption; Hydrogen desorption

Nomenclature			
A, B	pressure coefficients	Greek symbols	
C_a	hydrogen absorption reaction rate coefficient (s^{-1})	ΔH	enthalpy of reaction ($J\ mol^{-1}$)
C_d	hydrogen desorption reaction rate coefficient (s^{-1})	ε	porosity
C_p	heat capacity ($J\ kg^{-1}\ K^{-1}$)	λ	thermal conductivity ($W\ m^{-1}\ K^{-1}$)
d_f	outer width of fin (mm)	μ	dynamic viscosity ($kg\ m^{-1}\ s^{-1}$)
d_p	PCM thickness (mm)	ρ	density ($kg\ m^{-3}$)
d_s	outer diameter of fin (mm)	Subscripts	
E_a	activation energy ($J\ mol^{-1}$)	0	initial
E_d	hydrogen desorption activation energy ($J\ mol^{-1}$)	a	absorption
f	liquid fraction	ave	average
H	shell container height, (mm)	d	desorption
h	MH bed height (mm)	eff	effective
K	permeability (m^2)	emp	empty
k	mean saturated rate ($g\ s^{-1}$)	eq	equilibrium
L	latent heat of the PCM ($kJ\ kg^{-1}$)	g	gas
l_f	inner width of fin (mm)	in	inlet
M	molar mass ($kg\ kmol^{-1}$)	l	liquid
$\dot{m}$	hydrogen mass per unit time and unit volume ($kg\ m^{-3}\ s^{-1}$)	m	melting
m	mass (kg)	p	phase change material
$\vec{n}$	normal vector	ref	reference
P	pressure (MPa)	s	solid
R	universal gas constant ($J\ mol^{-1}\ K^{-1}$)	sat	saturated
T	temperature (K)	sol	solidification
t	time (s)	Abbreviation	
t_f	fin thickness (mm)	$KPIs$	key parameter indicators
V	Volume (m^3)	MH	metal hydride
$\vec{v}$	velocity in the metal hydride ($m\ s^{-1}$)	PCM	phase change material

<i>wt.%</i>	weight percentage	<i>SGSD</i>	system gravimetric storage density
<i>X</i>	reacted fraction	<i>SVSD</i>	system volumetric storage density

1 Introduction

The development of hydrogen energy and the achievement of an effective hydrogen economy are currently challenged by the available solutions for H₂ storage and transportation. Safety, large capacity, low cost and convenient access are the requirements for hydrogen storage technology. Metal hydride (MH), as a solid hydrogen storage technology, is considered as a potential candidate to meet the demand for hydrogen energy applications due to important advantages, among which the high volume hydrogen storage capacity, good safety and no explosion risks. However, during the hydrogen absorption/desorption cycle, there is always a large amount of reaction heat, which can easily cause the hydrogen absorption/desorption reaction to be suspended [1]. For example, the hydrogen absorption process is exothermic, which inevitably leads to an increase in the temperature of the MH bed along with an increase in the equilibrium pressure. Once the equilibrium pressure is too high to drive the reaction, the hydrogen absorption process stops, resulting in a rapid reduction in hydrogenation rate and a loss of H₂ storage capacity. Therefore, the improvement and accurate prediction of heat transfer plays a pivotal role in the design and optimization of metal hydride reactors to achieve high hydrogen storage efficiency [2].

Many researchers have analyzed the heat and mass transfer characteristics of metal hydride reactors by establishing mathematical models and experimental studies. Wu et al. [3] proposed a three-dimensional model of a new magnesium based metal hydride reactor with spiral coil heat exchanger. Based on the numerical simulation, the results showed that the spiral coil reactor could effectively improve the hydrogen storage rate. Bao et al. [4] evaluated the impact of two heat transfer enhancement measures, namely, using metal fins and making metal hydride powder into compacts. It was concluded that expanded natural graphite compression technology should be recommended in MHTES applications, while also removing expensive metal fins. Wang et al. [5] proposed a radiation microchannel reactor (RMCR) to improve thermal efficiency and developed a radiation microchannel reactor with a jacket (RMCR-J) to eliminate adverse heat transfer regions inside the RMCR. Compared with different forms of reactors, radiation tubes could effectively remove reaction heat and rapidly reduce the temperature of the MH bed, thereby accelerating the H₂ absorption process. Liu et al. [6] proposed a metal hydride reactor with variable cross-section annular fins (VCSAF). A thermal coupling model between the bed and VCSAF was established, and the influence of different fin structural parameters on the homogeneity of the reactor was studied. The VCSAF structure had good performance in eliminating non-uniform reaction. With the

expansion of reaction scale, the heat generation capacity had significantly improved, which had guiding significance for the engineering application of high temperature heat storage technology of MH. Bai et al. [7] proposed a metal hydride reactor with coupled fins and metal foam, and used a three-dimensional multi-physical model for research. The results showed that compared with the original metal foam reactor and fin reactor, the time required for the optimized reactor to complete 90% hydrogen charging was reduced by 6.9% and 38%, respectively. Under the conditions of different materials with high thermal conductivity, the optimal fin volume ratio of the optimized fin metal foam reactor is about 0.4.

The above researches mainly focus on innovation in the forms of heat exchange tubes and fins. In addition, some scholars have studied the combination of PCM to further improve the hydrogen absorption and desorption rates, as shown in Table 1.

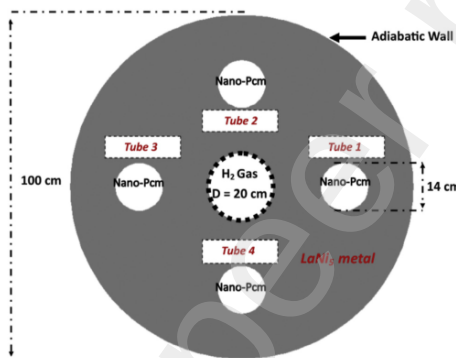


Figure 1 PCM is added in tubes of the metal hydride reactor [15]

Table 1 Summary of literature on metal hydride reactors with phase change materials

Ref.	Structure	KPIs	Considered parameters	Key findings
[8]	Metal hydride reactor with PCM jacket and coiled-tube heat exchanger	Hydrogen mass stored in metal hydride, and temperature distributions	1. Design of the hybrid heat exchanger. 2. Heat transfer coefficient, heating water temperature and outlet pressure. 3. Relationships among dehydriding rate, heat transfer coefficient, heating water temperature and outlet pressure.	1. The coiled-tube heat exchanger as a form of active heat transfer method shows better performance than the PCM as a kind of passive heat transfer method. 2. The relationship between the average dehydriding rate and three operation parameters is obtained.
[9]	Without PCM, pure PCM, PCM and metal foam reactors	Hydrogen mass stored in metal hydride, and temperature distributions	1. Different controlled flowrates. 2. Different hydrogen pressures. 3. The reactor's chargeability.	1. By embedding copper foam into the PCM, PCM-MF composite has a significant improvement on both thermal and charging/discharging performances of the MH canister. 2. The MH-PCM-MF system can maintain the charging /discharging for a significantly longer period compared to those achieved by using pure PCM.
[10]	Reactors with a finned tube heat exchanger and PCM	Reacted fraction and temperature distributions	1. The amount of PCM. 2. The inlet velocity of HTF. 3. The thermal conductivity of MH beds. 4. The hydrogen supply pressure.	1. By comparing with the reactor without PCM (MH-HTF reactor), the MH-HTF-PCM reactor's absorption time is shortened by 50%. 2. For the MH-HTF-PCM reactor, there are optimal amount of PCM and the inlet velocity of HTF, and increase hydrogen supply pressure can effectively accelerate the hydrogen

				absorption rate .
[11]	A novel MH-PCM unit (PCM sandwiched between two layers of MH)	Reacted fraction and temperature distributions	1. Mass ratio of PCM to MH (m_r). 2. The hydrogen supply pressure.	1. The decrease of m_r accelerates the desorption process. 2. When P_a is constant, there is a minimum mass ratio of PCM to MH ($m_{r,min}$) to completely absorb hydrogen. In order to fully absorb the hydrogen, P_a must increase to compensate for reduced m_r. 3. With the increase of P_a, $m_{r,min}$ and its change rate decrease.
[12]	Porous metal hydride tank (PMHT) with PCM and spiral tubes	Reacted fraction and temperature distributions, and PCM liquid fraction	1. The number of flat spiral tube planes (FSTPs). 2. The PCM-jacket thicknesses (annulus-shaped PCM and conical-shaped PCM).	1. Increasing the number of FSTPs throughout the PMHT affects the absorption duration by an 8% improvement. 2. The total hydrogen absorption duration is improved by 44% and 45% when using annulus-shaped PCMs and conical-shaped PCMs compared to the non-PCM PMHTs.
[13]	Metal Hydride reactor equipped with nano-enhanced PCM and U-tube heat exchanger.	Reacted fraction and PCM liquid fraction	1. The use of U-tube. 2. Add nanoparticles to the HTF. 3. The nano-enhanced PCM.	1. The U-tube heat exchanger can greatly improve the hydriding time duration by 33.5% compared to that with a basic MH tank configuration. 2. The effect of the addition of nanoparticles to the HTF to enhance the hydriding process is found not much evident. 3. The time duration of the hydriding process is reduced by 11.55% and 17.21% when the nano-enhanced PCM is added with 5% of graphene and 5% of CuO nanoparticles, respectively.

[14]	A metal hydride based thermal energy storage system (MH-TES) for the application of concentrated solar power system	Reacted fraction and temperature distributions, and heat flux	1. The MH thermal conductivity. 2. Fluid velocity. 3. The hydrogen pressure.	1. Increasing the thermal conductivity of the MH bed by adding expended graphite is beneficial to the heat transfer from the MH bed to heat transfer fluid. 2. the fluid velocity should match well with the thermal conductivity of the MH bed to keep the uniform heat exchange. 3. In the MH-TES system, reheating cycle contributes to increase the turbine work and reduces the temperature of the condenser, thus increasing the efficiency to 30.8%.
[15]	PCM is added in tubes of the metal hydride reactor (as shown in Figure 1)	Reacted fraction, temperature distributions, PCM liquid fraction, and energy for the different nano-PCM matrices	1. Design of the MH-nano-PCM system. 2. Heat transfer and solidification processes in the nano-PCM tubes.	1. The maximum energy loss is $1.39 \times 10^7 \text{ J kg}^{-1}$ in the proposed reactor. 2. The nano-PCM approach is a good strategy for latent energy via increasing the thermal conductivity of the PCM matrix. 3. The tube position greatly influenced the PCM solidification rate.
[16]	Metal Hydride reactor equipped with PCM	Nu number, temperature evolution	1. Nu numbers for different values of Ra. 2. Temperature variation.	1. The proposed method (a hybrid Lattice Boltzmann Phase-Field algorithm) can accurately and reliably describe the melting process of PCM.

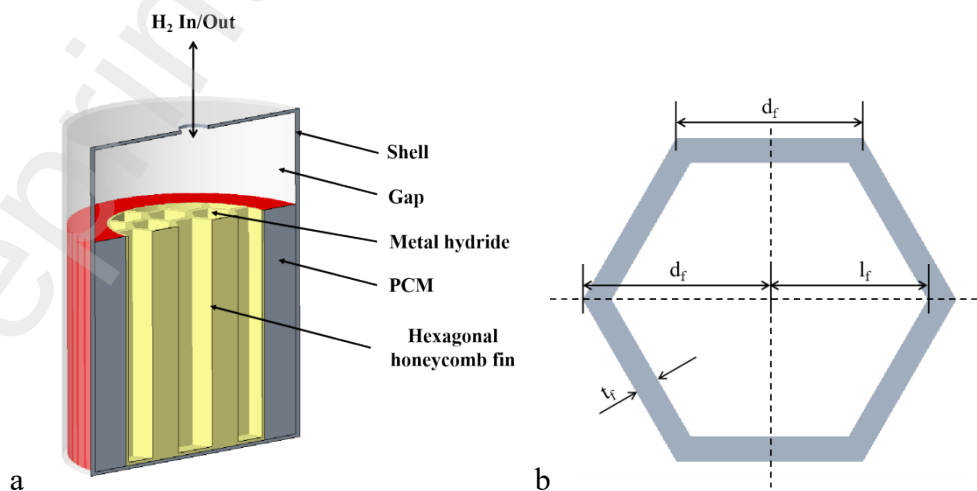
In the above, the studied KPIs and the considered parameters have been summarized and discussed. It can be found that the optimization of fin form has not been studied, and few scholars have conducted a systematic and comprehensive analysis of the KPIs proposed in this article for evaluating the reaction process.

In this paper, a new type of metal hydride reactor with honeycomb fins and integrated with phase change material is proposed to achieve uniform reaction and save time. The honeycomb structure has high structural efficiency and large surface area to volume ratio, resulting in good heat transfer performance. For this type of reactor, honeycomb fins are placed inside to increase the heat transfer area, metal hydride is filled into the space between the fins, and phase change material is surrounded outside. A three-dimensional reactor model has been built using the COMSOL Multiphysics V5.6 to predict the reaction process. The influence of the reactor with or without fins on the hydrogenation process is compared by numerical simulation, and the influence of the fin size, the PCM thickness, the PCM thermal conductivity and latent heat value, the porosity of MH and the initial temperature and pressure on the reactor performance is studied, so as to optimize the reaction parameters of the new structure. The results of this study can provide reference for using honeycomb fins and PCM as an effective thermal management method in MH storage systems.

2 Models, methods, and validation

2.1 Metal hydride reactor geometry and material properties

Figure 2 shows a schematic diagram of the MH reactor proposed in this study. The reactor is covered by a shell (cylinder), with an opening at the top for hydrogen to enter and exit. Hexagonal honeycomb fins are placed inside the reactor, surrounded by phase change material and filled with metal hydride. The dimensions of the reactor, including the honeycomb fins and PCM, are listed in Table 2.



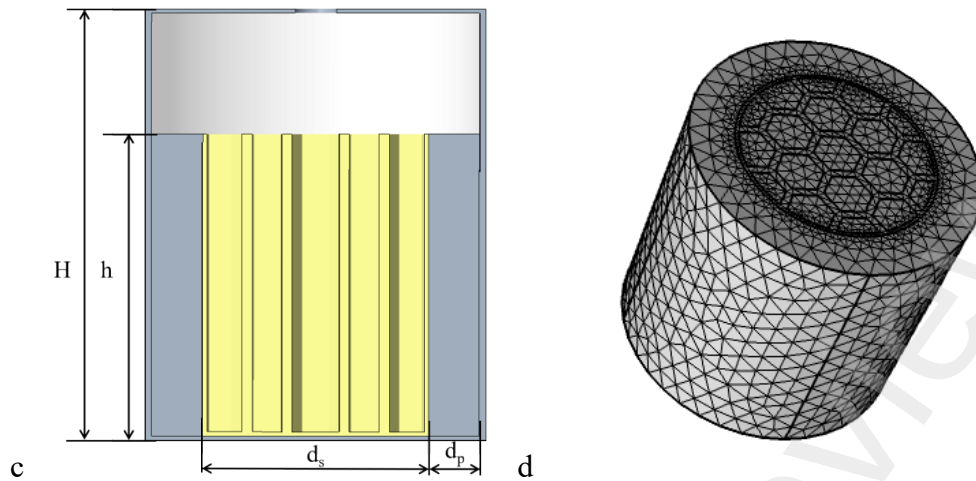


Figure 2 Schematic diagram of the metal hydride reactor combined with honeycomb fins and phase change material (a: 3D cross sectional view; b: single honeycomb fin cross sectional view; c: 3D cross sectional front view; d the calculation domain of the reactor with grids)

Table 2 Dimensions of honeycomb fins and PCM

Parameters	Value
Outer width of fin (variable value), d_f (mm)	7.5
Inner width of fin (variable value), l_f (mm)	6.35
Fin thickness (variable value), t_f (mm)	1
Outer diameter of fin (fixed value), d_s (mm)	52.5
PCM thickness (variable value), d_p (mm)	11.75
MH bed height (fixed value), h (mm)	70
Shell container height (fixed value), H (mm)	100

In the calculation domain, as shown in Figure 2d, the shell thickness is not considered, the reactor is insulated, and the upper gap is designed to be large enough, so only the inside of the reactor is simulated. In this study, LaNi_5 is selected as a hydrogen storage material and its thermophysical properties are shown in Table A.1 of Appendix A. It is crucial to select a suitable PCM to store and release heat during the hydrogen absorption and desorption process, PCM should have an appropriate melting temperature and latent heat, so that the reaction can proceed quickly. Therefore, $\text{LiNO}_3 \cdot 3\text{H}_2\text{O}$ is selected as the phase change material, with a phase change temperature of 303K and a latent heat of 296kJ kg^{-1} . Its specific thermophysical properties are shown in Table A.2 of Appendix A.

In order to develop and simplify the model describing the hydrogen absorption and desorption process, several assumptions are considered to establish the mathematical model for heat and mass transfer in MH and PCM:

- (1) Local thermal equilibrium is valid between H₂ and MH bed [17].
- (2) H₂ is an ideal gas [18].
- (3) Natural convection [18] and thermal radiation in MH are ignored [17].
- (4) MH is isotropic and has a uniform porosity [19].
- (5) Heat transfer through phase change media only occurs through conduction, and due to the weak melting front velocity of PCM, convection can be ignored [20].
- (6) The thermophysical properties of MH and PCM remain unchanged during the absorption process [3].
- (7) MH-PCM system is adiabatic [19].

2.2 Mathematical model of metal hydride reaction bed

2.2.1 Energy equation

The energy equation for MH bed is [20]:

$$(\rho C_p)_{eff,MH} \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_{eff,MH} \nabla T) + \dot{m} [\Delta H + T(C_{p,g} - C_{p,s})] \quad (1)$$

Where: ΔH is reaction enthalpy (J mol⁻¹); T is temperature (K); $C_{p,g}$ and $C_{p,s}$ are the specific heats of hydrogen and metal hydride, respectively (J kg⁻¹ K⁻¹); $(\rho C_p)_{eff,MH}$ represents the effective volumetric specific heat of MH; $\lambda_{eff,MH}$ represents the effective thermal conductivity of MH (W m⁻¹ K⁻¹), which can be calculated using equations (2) and (3).

$$(\rho C_p)_{eff,MH} = \varepsilon \rho_g C_{p,g} + (1 - \varepsilon) \rho_s C_{p,s} \quad (2)$$

$$\lambda_{eff,MH} = \varepsilon \lambda_g + (1 - \varepsilon) \lambda_s \quad (3)$$

Where: ε , ρ_g , ρ_s , λ_g , λ_s are porosity, H₂ density (kg m⁻³), MH density (kg m⁻³), H₂ thermal conductivity (W m⁻¹ K⁻¹), and MH thermal conductivity (W m⁻¹ K⁻¹), respectively.

2.2.2 Mass conservation equation

Solve the continuity equation in the MH layer to find the pressure and velocity fields of hydrogen in these regions, expressed as [21]:

$$\frac{\partial}{\partial t} (\varepsilon \rho_g) + \nabla \cdot (\rho_g \vec{v}) = -\dot{m} \quad (4)$$

$$\vec{v} = -\frac{K}{\mu} \nabla P \quad (5)$$

The mass conservation equation of MH bed is shown in equation (6):

$$(1 - \varepsilon) \frac{\partial \rho_s}{\partial t} = \dot{m} \quad (6)$$

Where: $\vec{v}$ is the Darcy velocity (m s^{-1}); $\dot{m}$ is the mass of hydrogen absorbed per unit time and volume ($\text{kg m}^{-3} \text{s}^{-1}$); K is the permeability inside the MH bed (m^2); μ is the H_2 dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$); ∇P is the pressure gradient (Pa).

2.2.3 Reaction kinetic energy

We compute the mass of absorbed and desorbed hydrogen per unit time and volume as proposed in [22]:

- absorption:

$$\dot{m} = C_a \exp\left(-\frac{E_a}{RT}\right) \ln\left(\frac{P_g}{P_{eq}}\right) (\rho_{sat} - \rho_s) \quad (7)$$

- desorption:

$$\dot{m} = C_d \exp\left(-\frac{E_d}{RT}\right) \left(\frac{P_g - P_{d,eq}}{P_{d,eq}}\right) (\rho_s - \rho_0) \quad (8)$$

Where: C_a and C_d are the rates of hydrogen absorption and desorption reaction, respectively (s^{-1}); E_a and E_d are the activation energies for hydrogen absorption and desorption, respectively (J mol^{-1}); ρ_{sat} is the saturated density of metal hydride (kg m^{-3}), ρ_s is the density of the metal hydride reaction process (kg m^{-3}), ρ_0 is the initial density of metal hydride (kg m^{-3}). Table A.2 of Appendix A reports all the values employed for the described parameters. P_g is the hydrogen pressure (MPa); P_{eq} is the equilibrium pressure in [MPa] of the reaction process, which can be calculated in the general form of the Van't-Hoff equation [23], as shown in equations (9) and (10).

$$\ln\left(\frac{P_{a,eq}}{P_{ref}}\right) = A_a - \frac{B_a}{T} \quad (9)$$

$$\ln\left(\frac{P_{d,eq}}{P_{ref}}\right) = A_d - \frac{B_d}{T} \quad (10)$$

In the above, when the reference pressure P_{ref} is 1MPa, the values of A and B in hydrogen absorption and desorption processes are $A_a=10.7$ $B_a=3704.6$, $A_d=10.57$, $B_d=3704.6$, respectively.

The kinetics equations are formulated as follows [5]:

- absorption:

$$\left(\frac{dX}{dT}\right)_a = C_a \exp\left(-\frac{E_a}{RT}\right) \ln\left(\frac{P_g}{P_{eq}}\right) (1 - X) \quad (11)$$

- desorption:

$$\left(\frac{dX}{dT}\right)_d = C_d \exp\left(-\frac{E_d}{RT}\right) \left(\frac{P_g - P_{d,eq}}{P_{d,eq}}\right) X \quad (12)$$

Where: X is the reacted fraction.

The storage capacity for hydrogen is generally measured with regard to weight percentage, the calculated value of hydrogen storage in this paper is the hydrogen storage capacity within 8000s and the calculation formula is given as [24]:

$$wt.\% = \frac{\text{mass of hydrogen absorbed}}{\text{mass of alloy used}} = \frac{100(\rho_s - \rho_0)}{\rho_0} \quad (13)$$

2.3 PCM mathematical model

For PCM, the liquid phase change material has high viscosity, so the convective heat transfer can be ignored [21]. Therefore, the energy equation for PCM is as follows.

$$(\rho C_p)_{eff,PCM} \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_{eff,PCM} \nabla T) \quad (14)$$

Where: $(\rho C_p)_{eff,PCM}$ represents the effective volumetric specific heat of PCM; $\lambda_{eff,PCM}$ represents the effective thermal conductivity of PCM ($\text{W m}^{-1} \text{K}^{-1}$), the calculation formula is as follows:

$$(\rho C_p)_{eff,PCM} = f \rho_{p,l} C_{p,p,l} + (1-f) \rho_{p,s} C_{p,p,s} + \rho_p L \frac{\partial \alpha}{\partial T} \quad (15)$$

$$\lambda_{eff,PCM} = f \lambda_{p,l} + (1-f) \lambda_{p,s} \quad (16)$$

Where: $\rho_{p,l}$ and $\rho_{p,s}$ are the density of the liquid and solid phases of PCM, respectively (kg m^{-3}); $C_{p,p,l}$ and $C_{p,p,s}$ are the specific heat of the liquid and solid phases of PCM ($\text{J kg}^{-1} \text{K}^{-1}$); f is the liquid fraction of PCM, which is calculated according to equation (17) [20]; L is the latent heat of PCM (kJ kg^{-1}). The variable density ρ_p and α of PCM are calculated by equations (18) and (19).

$$f = \begin{cases} 1 & T \geq T_{mel} \\ \frac{(T - T_{sol})}{(T_{mel} - T_{sol})} & T_{sol} < T < T_{mel} \\ 0 & T \leq T_{sol} \end{cases} \quad (17)$$

$$\rho_p = (1-f) \rho_{p,s} + f \rho_{p,l} \quad (18)$$

$$\alpha = \frac{1}{2} \frac{(1-f) \rho_{p,s} - f \rho_{p,l}}{\rho_p} \quad (19)$$

Where: T_{mel} and T_{sol} are the melting and solidification temperatures of PCM, respectively (K).

2.4 Initial and boundary conditions

2.4.1 Initial conditions

For the hydrogen absorption process, the initial temperature of MH and PCM is set as 293.15K, and the initial MH density is taken as the hydrogen free metal density, which is 7164 kg m⁻³. The hydrogen pressure is 0.8MPa, and the initial MH pressure is the equilibrium pressure at the temperature of 293.15K. The MH pressure changes linearly from the initial equilibrium pressure to 0.8MPa within 100s, and then remains constant during the subsequent absorption process. For the hydrogen desorption process, the initial temperature is set as 313.15K, the initial MH density is 7259 kg m⁻³, and the hydrogen pressure is 0.1MPa. The MH pressure changes linearly from the initial equilibrium pressure to 0.1MPa within 100s.

For hydriding:

$$T(r, \theta, z, 0) = T_0 = 293.15K \quad (20)$$

$$\rho_s(r, \theta, z, 0) = \rho_0 = \rho_{emp} \quad (21)$$

$$P_g(r, \theta, z, 0) = P_0 = 0.8MPa \quad (22)$$

$$P_{in} = \begin{cases} P_{a,eq}(T_0) + [P_0 - P_{a,eq}(T_0)] \times t / 100 & 0 < t < 100s \\ P_0 & t \geq 100s \end{cases} \quad (23)$$

For dehydriding:

$$T(r, \theta, z, 0) = T_0 = 313.15K \quad (24)$$

$$\rho_s(r, \theta, z, 0) = \rho_0 = \rho_{sat} \quad (25)$$

$$P_g(r, \theta, z, 0) = P_0 = 0.1MPa \quad (26)$$

$$P_{in} = \begin{cases} P_{d,eq}(T_0) + [P_0 - P_{d,eq}(T_0)] \times t / 100 & 0 < t < 100s \\ P_0 & t \geq 100s \end{cases} \quad (27)$$

2.4.2 Boundary conditions

The boundary conditions are as follows:

Inlet boundary condition [25]:

$$P_g(r, \theta, z, 0) = P_{in} \quad (28)$$

$$\nabla T \cdot \vec{n} = 0 \quad (29)$$

The conjugate heat transfer boundary condition between MH bed and PCM is given by:

$$\lambda_{MH} \nabla T_{MH} \cdot \vec{n} = \lambda_{PCM} \nabla T_{PCM} \cdot \vec{n} \quad (30)$$

2.5 Key performance indicators (KPIs)

2.5.1 System gravimetric and volumetric storage density (SGSD and SVSD)

The system gravimetric and volumetric storage densities of the developed MH reactor are calculated using the following equations [26]. The total mass of the reactor includes the mass of metal hydride, the mass of fins and the mass of phase change material. The total volume of the reactor includes the volume of metal hydride, the volume of fins and the volume of phase change material.

$$\text{System gravimetric storage density} = \frac{\text{mass of the hydrogen stored}}{\text{total mass of the reactor}} \quad (31)$$

$$\text{System volumetric storage density} = \frac{\text{mass of the hydrogen stored}}{\text{total volume occupied by the reactor}} \quad (32)$$

2.5.2 Mean saturated rate

The mean hydrogen absorption and desorption rates are calculated as follows [7]:

$$k = \frac{m_{H_2, sat}}{t} \quad (33)$$

2.5.3 Average MH temperature

The change of average temperature curve can also reflect the reaction rate. The average MH temperature in the calculation domain is calculated as follows:

$$T_{ave} = \frac{\int_{V_{MH}} T dV_{MH}}{V_{MH}} \quad (34)$$

Where: V_{MH} is the volume of MH.

2.6 Grid independence verification

Figure 3 shows the grid independence verification of the reactor mathematical model. The model is tested using four different tetrahedral grids characterized by 72171, 124701, 203297, and 294827 tetrahedral elements, respectively. The specific allocation of grid quantity is shown in Table 3. As can be seen from figure 3, there is no significant difference in temperature and reaction fraction between Grid 3 and Grid 4. Therefore, while ensuring computational accuracy, in order to save time and cost, the grid with 203297 elements is selected for the following.

Table 3 Grid parameters

Number	Number	Number	Edge	Vertex	Average	Unit	Grid
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	of tetrahedral elements	of triangular elements	element	unit	unit mass	volume ratio	volume (m ³)
1	72171	17792	2572	208	0.6019	0.002022	3.152×10^{-4}
2	124701	26003	3086	208	0.6319	0.002274	3.162×10^{-4}
3	203297	38429	3708	208	0.6575	0.00259	3.166×10^{-4}
4	294827	49811	4393	208	0.6687	0.00255	3.171×10^{-4}

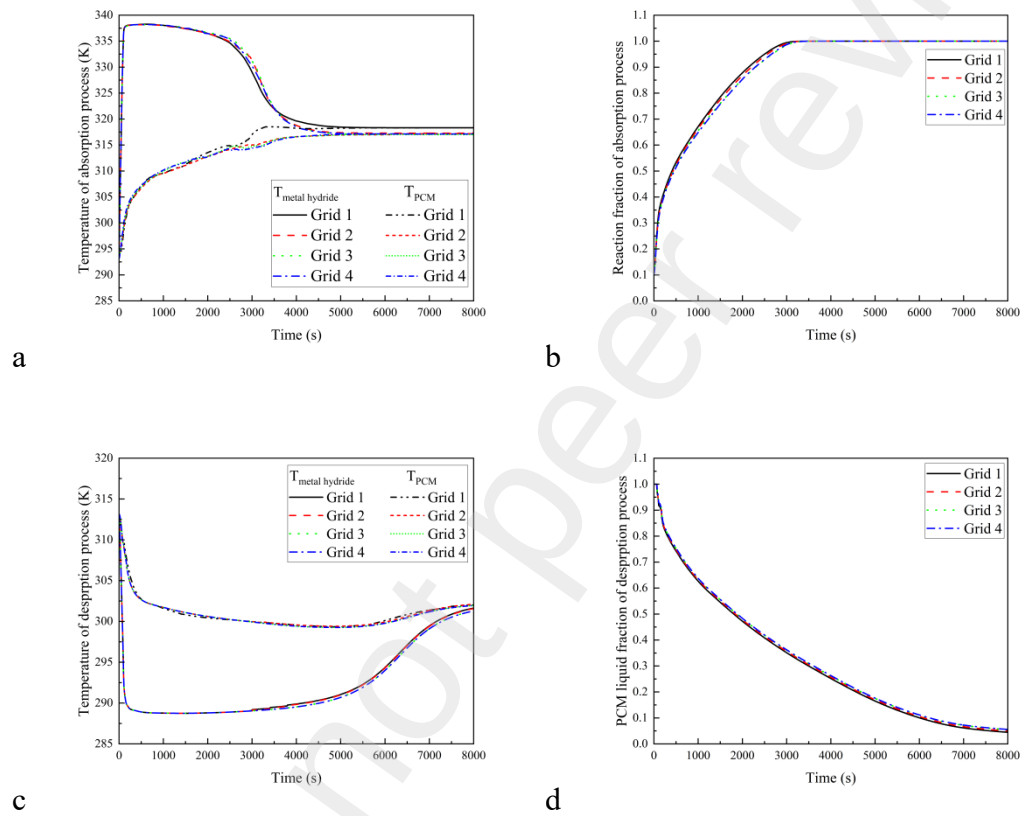


Figure 3 Grid independence assessment of the reactor meshes (a: comparison of MH temperature and PCM temperature in hydrogen absorption process; b: comparison of reaction fraction in hydrogen absorption process; c: comparison of MH temperature and PCM temperature in hydrogen desorption process; d: comparison of PCM liquid fraction in hydrogen desorption process)

2.7 Simulation methods and model validation

The above equations of MH bed and PCM are solved numerically using COMSOL Multiphysics V5.6 software. The time dependent solver is used to solve the change of reactor temperature distribution, hydrogen absorption and PCM liquid phase distribution with time [23]. The relative error and absolute error are set to 10^{-3} and 10^{-4} respectively.

In order to investigate the reliability of the current model, comparisons are made with the experimental and numerical results of Jemni et al. [27], Chung et al. [23], and Maad et al. [28]. Firstly, in the absence of PCM, with pressure and temperature of 0.8MPa and 293K respectively, the comparison with the results of Jemni et al. [27] and Chung et al. [23] is shown in Figure 4. From the figure, it can be seen that the maximum deviation between numerical results and the experimental results is around 5%. Therefore, the established model proves effective in simulating the process of hydrogen absorption and desorption.

In order to verify the reliability of the model in the presence of the phase change material, the numerical results are compared with those of Maad et al. [28], as shown in Figure 5. The figure shows the variation of liquid fraction with time, with a maximum deviation between the two sets of results of about 4.3%. Therefore, our model provides reliable predictions also for PCM-enhanced reactors.

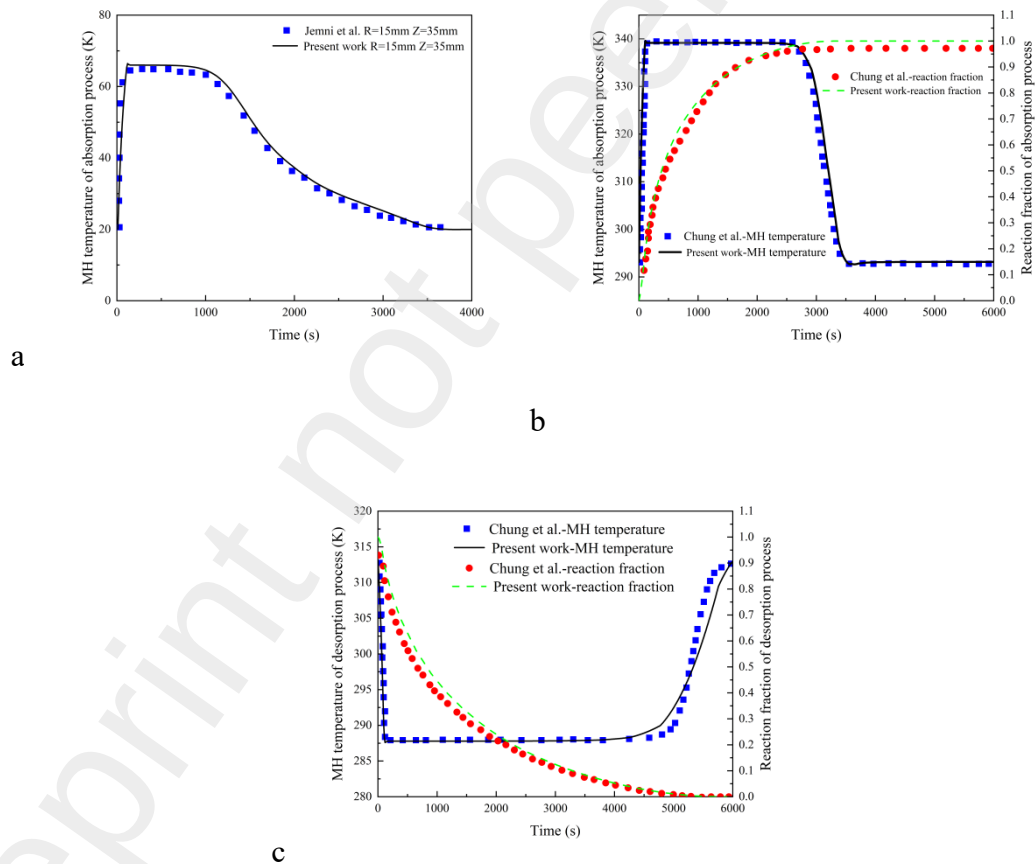


Figure 4 Comparison of simulation results with experimental and numerical data (a: comparison of MH temperature in hydrogen absorption process with Jemni et al. [27]; b: comparison of MH temperature and reaction fraction in hydrogen absorption process with Chung et al. [23]; c: comparison of MH temperature and reaction fraction in hydrogen desorption process

with Chung et al. [23])

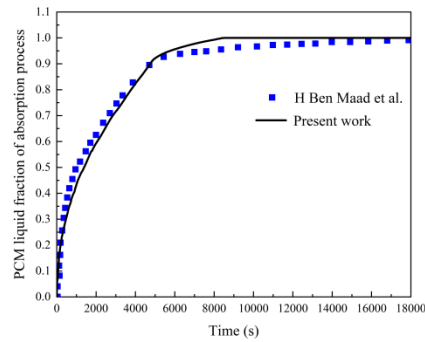


Figure 5 Comparison with the numerical results of liquid fraction of PCM obtained by Maad et al.

[28]

3 Results and discussion

In this section, the numerical simulation results of the hydrogen absorption and desorption process in the proposed honeycomb fin metal hydride reactor are presented. The effects of different geometrical conditions (fin size, PCM thickness) and operating conditions (thermal conductivity of PCM, initial temperature and pressure) on the hydrogen absorption and desorption reaction are investigated.

3.1 Hydrogen absorption process

3.1.1 Performance comparison of reactors with and without fins

When hydrogen is injected at 0.8MPa into the reactor, the pressure is greater than the equilibrium pressure inside the reactor, leading to hydrogen absorption in the MH and heat release. By controlling the same mass of filled metal hydride and phase change material, a metal hydride reactor with hexagonal honeycomb fins and a metal hydride reactor without fins are compared at 0.8MPa and 293.15K. The changes in temperature, reaction fraction, and liquid phase fraction over time in the reactor are shown in Figure 6. The reactor with hexagonal honeycomb fins achieved a reaction rate of 90% at 2420s, which is more than 66% higher than the reactor without fins at 7090s. At the initial stage of the reaction, the reaction fraction of the two is similar to the bed temperature. After reaching its maximum, the temperature begins to decrease. As the reaction continues, the proposed hexagonal honeycomb fin reactor exhibits a faster hydrogen absorption rate compared to the reactor without fins. This is because at the initial stage of the reaction, hydrogen enters the reactor and generates heat: at this stage the absorption rate is weakly dependent on the heat transfer from the alloy bed and the reaction kinetic plays a more important role in determining the reaction rate, compared to heat transfer to the bed [25]. As the reaction proceeds, heat transfer becomes more

important. The heat generated by hydrogen absorption is released in a short time and transferred to the phase change material. After reaching the phase change temperature, the phase change material stores the heat released by the reaction, and the metal hydride temperature gradually decreases until equilibrium. Due to the heat transferred from the outermost part of the reactor to the phase change material, the internal heat transfer of the reactor without fins is slow, resulting in a slow drop in metal hydride temperature and a slow rise in PCM temperature. Finally, the phase transition of PCM completely occurs and the reaction rate slows down. The relevant KPIs are shown in Table 4 of Section 3.1.5: it can be seen that the adoption of hexagonal honeycomb fins improves the heat transfer and hydrogen absorption rate. However, the Table shows that due to the presence of the fins, the overall gravimetric and volumetric storage densities of the reactor are reduced.

In order to observe the heat transfer phenomenon in the reactor more intuitively, the changes in temperature and reaction fraction at several points in time in the reactor are shown in Figure B.1 of Appendix B.

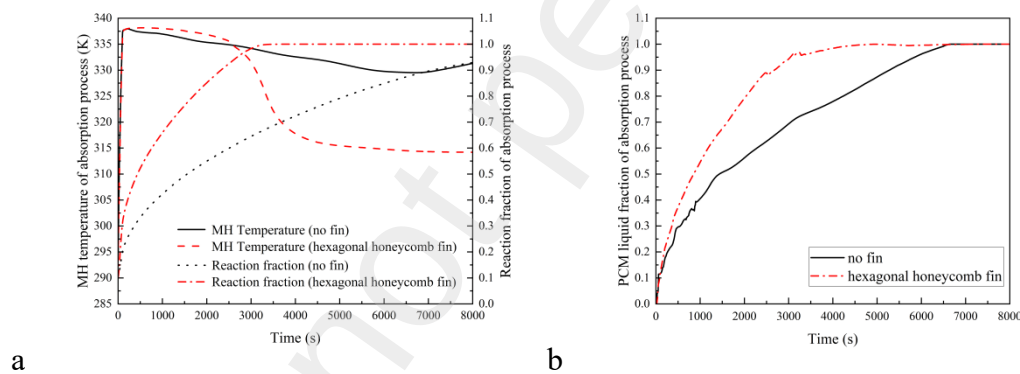


Figure 6 Comparison of (a) temperature and reaction fraction; (b) liquid fraction of PCM in the reactors with and without fins

3.1.2 Effect of fin size on hydrogen storage

On the premise that the volume of metal hydride and phase change material is basically the same, the width d_f and thickness t_f of hexagonal honeycomb fin are changed to 6.5mm/0.86mm, 7.5mm/1mm, 8.5mm/1mm, respectively. The physical parameters are reported in Table 5: the bold values represent the initial parameters. The reaction fraction, bed temperature, and liquid fraction of three different fin sizes are compared at 0.8MPa and 293.15K, as shown in Figure 7. The results show that the reaction time of 6.5 mm/0.86 mm fin size is less, and the hydrogen absorption rate and temperature drop faster. The specific numerical changes are shown in Table 4 of

Section 3.1.5. The mean saturated rate decreases from 1.269×10^{-3} for 6.5mm fins to 1.204×10^{-3} for 8.5mm fins. This is due to the higher fin density in the reactor with the 6.5mm fins, which makes the metal hydride distribution more uniform and the heat transfer faster, while the reaction evolution in the presence of 7.5 mm and 8.5 mm fins is basically the same. Compared with the two forms, the 6.5 mm fin reaction rate increases by about 7%, and the hydrogen storage capacity of the three fin forms all reach 1.326 wt.%. The specific change processes at several points in time in the reactor are shown in Figure B.2 of Appendix B.

Table 5 Parameters of different fin sizes

Parameters	Value
Fin width, d_f (mm)	6.5/7.5/8.5
Fin thickness, t_f (mm)	0.86/1.0/1.0
MH volume, V_{MH} (cm ³)	49.45/49.22/49.36
MH mass, m_{MH} (kg)	0.354/0.353/0.354

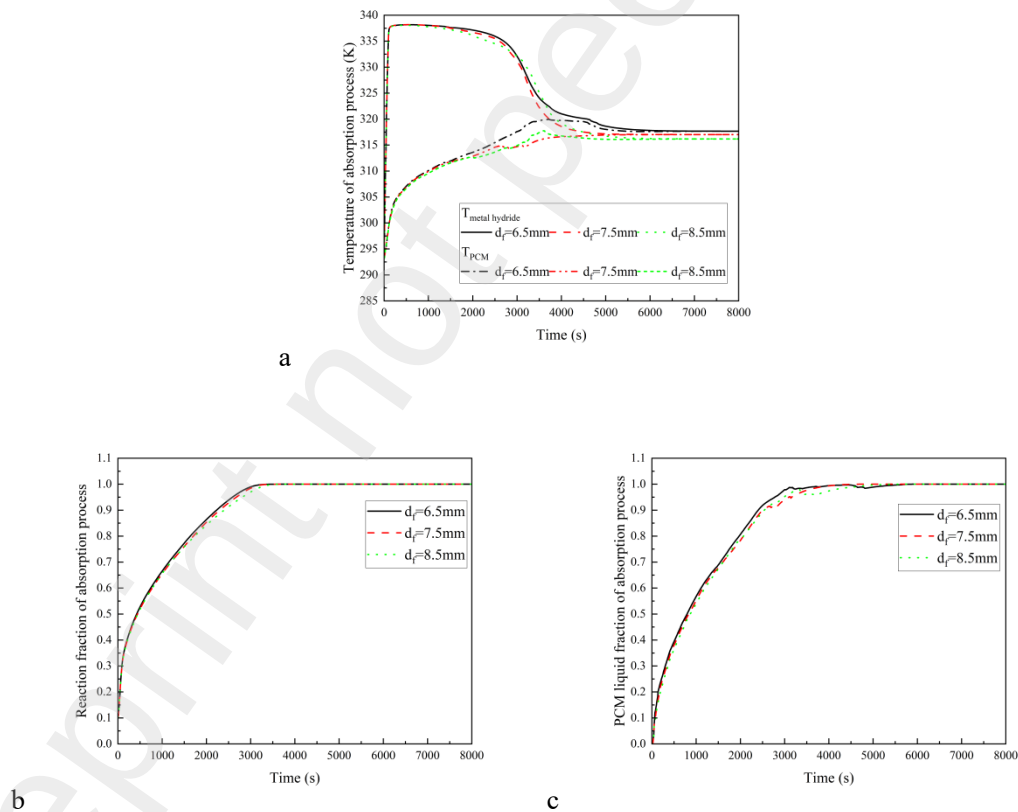


Figure 7 Under different fin sizes, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.1.3 Effect of changes in PCM thickness on hydrogen storage

In this section we evaluate the effect of PCM thickness on reactor performance.

The results are summarized in Figure 8. The PCM thickness d_p is 11.25mm, 11.75mm, 12.25mm, and 12.75mm, respectively. The specific volume and mass parameters of PCM are shown in Table 6, with the bold values in the table representing the initial parameters. It can be seen that when PCM thickness d_p increases from 11.25mm to 12.75mm, MH has a faster heat release and reaction rate, thereby improving its heat storage capacity. Due to the increase in volume and mass of PCM, it can absorb more reaction heat and increase the heat release rate of MH, so that the equilibrium temperature reached by bed temperature is lower. The time required for the reactor to reach 90% reaction fraction is 2480s, 2420s, 2360s, and 2300s, respectively. All reactors reach a hydrogen storage capacity of 1.326 wt.%, but at different times. The hydrogen absorption time is reduced by about 7% when the PCM thickness increases from 11.25mm to 12.75mm. Besides, the other KPIs are shown in Table 4. We find that, as the PCM thickness increases, the total mass and volume of the reactor grow, resulting in a decrease in the gravimetric and volumetric storage densities. Such a decrease is balanced by an increase in the mean saturated rates. By comparison, the gravimetric and volumetric storage densities in reference [26] are 0.73% and 20.4 kg of H₂ per m³, respectively, while the gravimetric and volumetric storage densities of our reactor are decreased by an average of 44.1%, 28.3%, respectively. This is because the application of honeycomb fins and PCM increases the volume and mass of the reactor. However, the adoption of honeycomb fins provides a faster more effective heat transfer, accelerating the reaction rates of hydrogen absorption and desorption. This enhancement is sided by the reduction in energy consumption due to the heat recovered through the exploitation of the PCM. The specific reaction fraction, temperature, and liquid fraction change processes at several points in time in the reactor are shown in Figure B.3 of Appendix B.

Table 6 Parameters of different PCM thickness

Parameters	Value
PCM thickness, d_p (mm)	11.25/ 11.75 /12.25/12.75
PCM volume, V_{PCM} (cm ³)	157.72/ 166.02 /174.43/182.95
PCM mass, m_{PCM} (kg)	0.309/ 0.325 /0.342/0.359

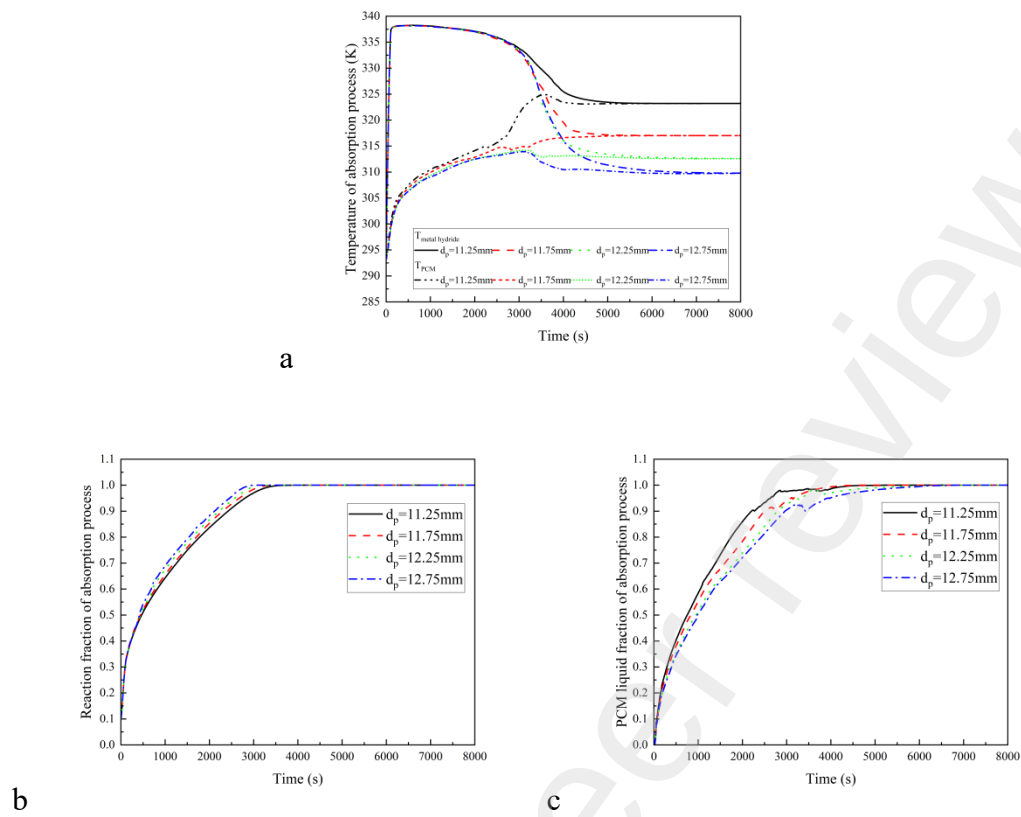


Figure 8 Under different PCM thickness, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.1.4 Effect of changes in operating parameters on hydrogen storage

3.1.4.1 Effect of solid thermal conductivity of PCM

The changes of bed temperature, PCM temperature, reaction fraction and liquid fraction with time during hydrogen absorption in metal hydride reactor under different solid thermal conductivity of PCM are shown in figure 9. The solid thermal conductivities are $1.32\text{W m}^{-1}\text{K}^{-1}$, $2.64\text{W m}^{-1}\text{K}^{-1}$, $5.28\text{W m}^{-1}\text{K}^{-1}$, $7.92\text{W m}^{-1}\text{K}^{-1}$, and $15.84\text{W m}^{-1}\text{K}^{-1}$, respectively, while the other parameters remain unchanged. As shown in the figure, when the solid thermal conductivity of PCM increases from $1.32\text{W m}^{-1}\text{K}^{-1}$ to $15.84\text{W m}^{-1}\text{K}^{-1}$, the time for hydrogen absorption decreases, and the time to reach 1.326 wt.% hydrogen storage is also shorter, and the hydrogen absorption time is reduced by about 20%. The corresponding mean saturated rates are $1.222 \times 10^{-3}\text{ g s}^{-1}$, $1.226 \times 10^{-3}\text{ g s}^{-1}$, $1.239 \times 10^{-3}\text{ g s}^{-1}$, $1.414 \times 10^{-3}\text{ g s}^{-1}$, and $1.554 \times 10^{-3}\text{ g s}^{-1}$. This is because the increase of the solid thermal conductivity allows a faster transfer and a larger storage in the PCM of the heat generated by hydrogen absorption process.

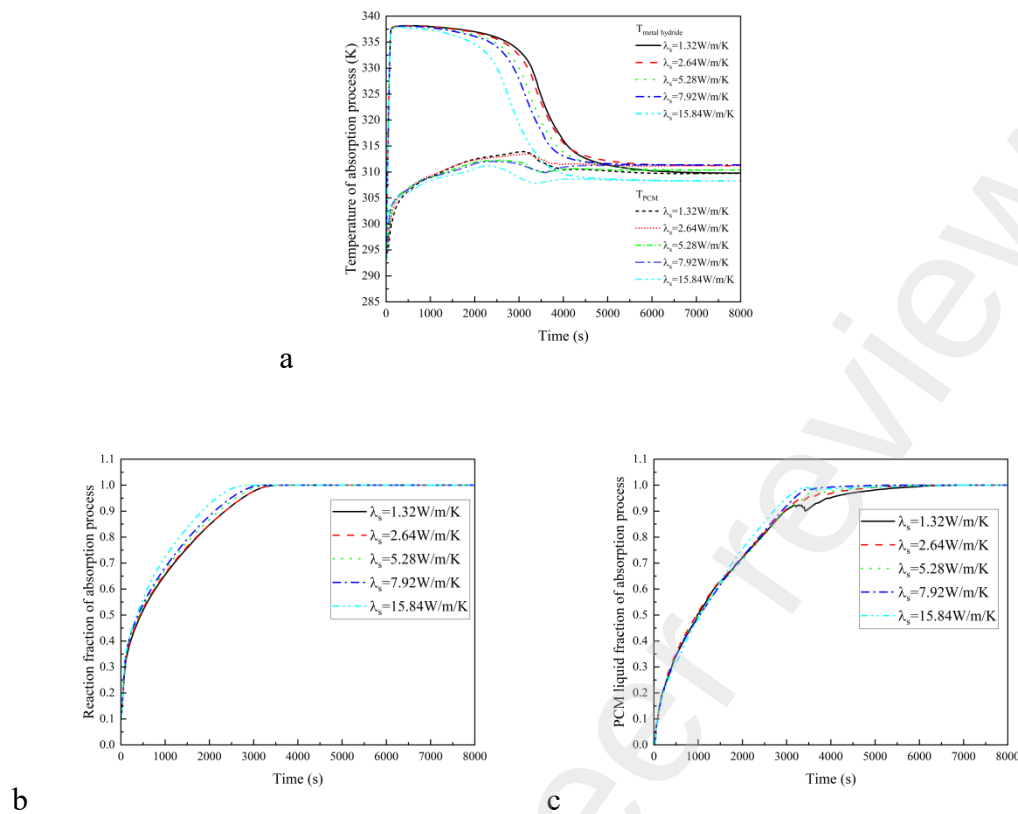


Figure 9 Under different solid thermal conductivity of PCM, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.1.4.2 Effect of liquid thermal conductivity of PCM

From the above, it can be seen that when the solid thermal conductivity increases by 12 times, the reaction time only decreases by 20%. This shows that thermal conductivity has little effect on the reactor bed temperature and hydrogen absorption reaction fraction. This may be due to the solid thermal conductivity of the PCM affecting only the initial stage of heat transfer [29]. When the phase change temperature is reached, as the reaction heat transfers, the PCM gradually undergoes a phase transition from solid to liquid from the inside out, and excess reaction heat is stored in the liquid PCM. As the liquid fraction of the PCM becomes larger, the impact of the solid PCM thermal conductivity on the reactor performance becomes negligible.

When the liquid thermal conductivity is $0.58 \text{ W m}^{-1} \text{ K}^{-1}$, $1.16 \text{ W m}^{-1} \text{ K}^{-1}$, $2.32 \text{ W m}^{-1} \text{ K}^{-1}$ and $3.48 \text{ W m}^{-1} \text{ K}^{-1}$ respectively, the bed and PCM temperatures, reaction fraction and liquid fraction in the metal hydride reactor are shown in Figure 10. From Figure 10, it can be seen that the time taken for the bed temperature to drop to the equilibrium temperature decreases by about 3100s, and the maximum temperature reached by the bed decreases. The time for the reactor to reach 90% reaction rate

decreases by about 71%. The corresponding KPIs are shown in Table 4. It can be seen that the mean saturated rate gradually increases. In addition, when the liquid thermal conductivity increases from $2.32 \text{ W m}^{-1} \text{ K}^{-1}$ to $3.48 \text{ W m}^{-1} \text{ K}^{-1}$, the hydrogen absorption rate increases slightly, thus further increasing the liquid thermal conductivity does not provide any significant improvements. This is because the heat diffusion can only reach the heat unsaturated PCM through the metal hydride bed, honeycomb fins and liquid phase change material. When the liquid thermal conductivity of PCM increases and exceeds the thermal conductivity of the metal hydride bed, the main limiting factor of heat diffusion changes from the liquid thermal conductivity of PCM to the thermal conductivity of the metal hydride bed.

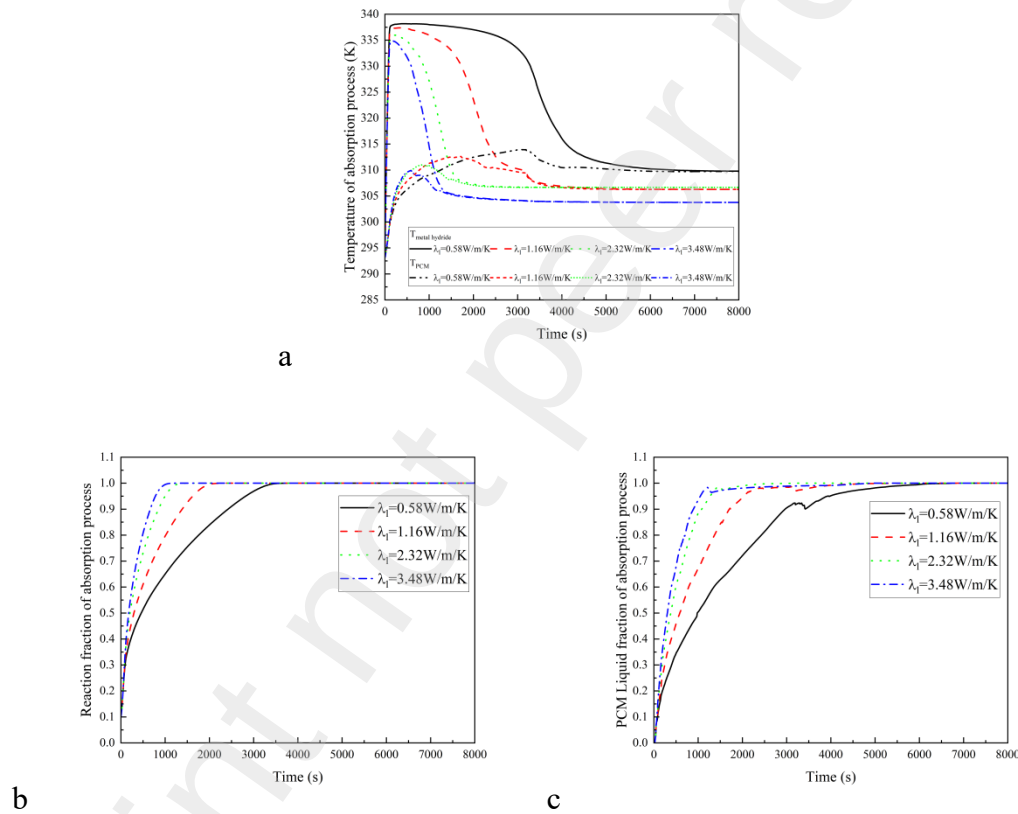
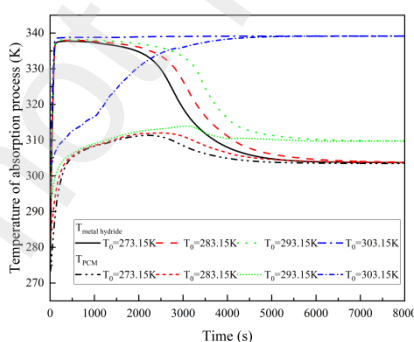


Figure 10 Under different liquid thermal conductivity of PCM, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

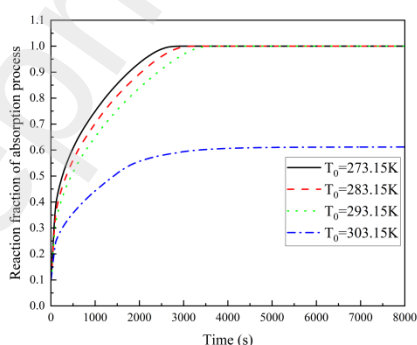
3.1.4.3 Effect of initial temperature

The effect of initial temperature on the hydrogen absorption process is studied, as shown in Figure 11. For each case, the initial temperature values are 273.15K, 283.15K, 293.15K, and 303.15K, respectively. All other parameters are left unchanged. The figure shows that changes in the initial temperature have a significant impact on the hydrogen absorption. Through the change of reaction fraction and bed temperature, it

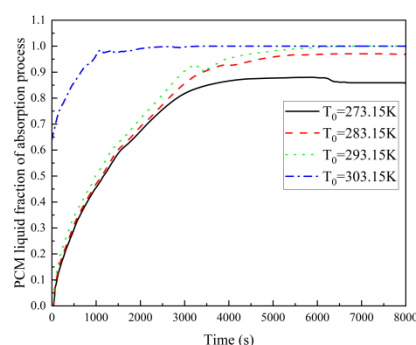
can be seen that lower initial temperature has faster reaction rate. At an initial temperature of 273.15K, approximately 75% of hydrogen is absorbed within 1000s, the mean saturated rate is $1.591 \times 10^{-3} \text{ g s}^{-1}$, while, at an initial temperature of 303.15K, the reaction fraction reaches only 0.61 even after 8000s, with a hydrogen storage capacity of 0.811 wt.%, and the mean saturated rate is $0.359 \times 10^{-3} \text{ g s}^{-1}$. This shows that the reaction rate decreases as the initial temperature increases. This can be ascribed to the fact that, since the melting temperature of PCM is 303K, when the initial temperature is set to 273.15K, the difference with the melting temperature of PCM is about 30K. The heat released by hydrogen absorption can be fully absorbed by PCM, promoting rapid reaction. However, when the initial temperature is 303.15K, the melting temperature of PCM has already been reached, causing PCM to undergo phase transition at the beginning of the reaction. This results in an initial temperature that is too high, causing the PCM temperature to rise too fast to fully absorb the heat generated by the reaction. As shown in the figure 11 of the bed temperature and liquid fraction, bed temperature can not be declined at high temperature, liquid phase fraction within 660s has reached 90%, so that the hydrogen absorption rate and the amount of hydrogen absorption are reduced. Therefore, the lower initial temperature can promote the process of reaction.



a



b



c

Figure 11 Under different initial temperature, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.1.4.4 Effect of hydrogen absorption pressure

In addition to the influencing factors studied above, hydrogen absorption pressure is also an important operating parameter of hydrogen storage reaction. Figure 12 shows the influence of hydrogen pressure (0.6MPa, 0.8MPa, 1.0MPa, 1.2MPa) on the absorption process. The results indicate that the increase in hydrogen pressure effectively improves the heat transfer and hydrogen absorption rate. During the hydrogen absorption process, the temperature first rises rapidly to the highest values corresponding to different pressures, then maintains for a period of time before rapidly decreasing, and then decreases at a slower rate. The sharp increase in temperature is due to the low temperature at the beginning of the reaction, low equilibrium hydrogen pressure, large reaction driving force (the difference between hydrogen pressure in the reactor and equilibrium hydrogen pressure), and fast reaction speed; the reaction fraction sharply increases, and in a short time, a large amount of heat is released from the reaction, which cannot be transmitted from the bed layer to the phase change material zone in time, resulting in a sharp increase in bed temperature. After reaching the highest temperature, the driving force of the reaction begins to decrease, and the reaction tends to be gentle. The heat generated by the reaction is slowly transferred to the phase change material, and the heat generated by the reaction is basically in equilibrium with the heat transferred to the PCM. Therefore, it will be maintained for a period of time at the highest temperature. Then the temperature rapidly decreases, and the temperature in the phase change zone rises to the phase change temperature. The PCM begins to melt, absorbing and storing the heat. Finally, as the reaction approaches completion and the temperature difference between the reaction zone and the phase change material gradually reduces, causing a decrease in the bed temperature, and causing a change of slope for the reaction and liquid fractions. In addition, it can be seen from the figure that the higher the hydrogen pressure, the greater the rise in temperature, leading to the highest temperatures of 330K, 338K, 345K, and 351K, respectively. The large pressure difference, in fact, provides the driving force in the reaction process, promoting the progress of the reaction. The time for the reactor to reach 90% reaction fraction is 3420s, 2300s, 1900s, and 1640s, respectively, the hydrogen absorption time is reduced by about 52%, and the corresponding mean saturated rates are $0.840 \times 10^{-3} \text{ g s}^{-1}$, $1.222 \times 10^{-3} \text{ g s}^{-1}$, $1.431 \times 10^{-3} \text{ g s}^{-1}$, and $1.700 \times 10^{-3} \text{ g s}^{-1}$.

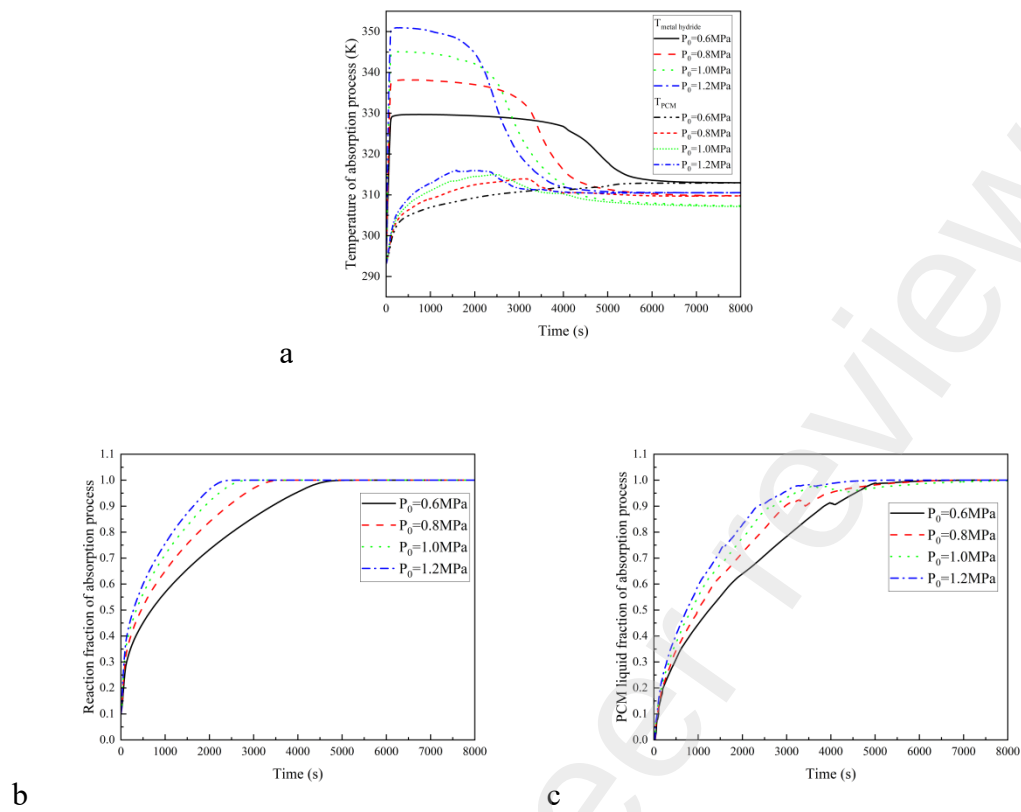


Figure 12 Under different hydrogen absorption pressure, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.1.5 Key performance indicators' results of hydrogen absorption process

The values of weight percentage, system gravimetric storage density, system volumetric storage density and mean saturated rate corresponding to the changes of different parameters during hydrogen absorption are shown in Table 4.

Table 4 Key performance indicators' results of hydrogen absorption process

Parameters	wt. %	SGSD (%)	SVSD (kg of H ₂ per m ³)	k (g s ⁻¹)
reactor without fin	1.232	0.642	16.52	0.545×10^{-3}
the developed MH reactor	1.326	0.411	14.76	1.222×10^{-3}
Different fin size (mm)				
6.5	1.326	0.411	14.76	1.269×10^{-3}
7.5	1.326	0.411	14.76	1.222×10^{-3}
8.5	1.326	0.411	14.76	1.204×10^{-3}
Different PCM thickness (mm)				
11.25	1.326	0.417	15.19	1.062×10^{-3}
11.75	1.326	0.411	14.76	1.222×10^{-3}

12.25	1.326	0.405	14.40	1.326×10^{-3}
12.75	1.326	0.399	14.05	1.549×10^{-3}
Different solid thermal conductivity of PCM ($\text{W m}^{-1} \text{K}^{-1}$)				
1.32	1.326	0.411	14.76	1.222×10^{-3}
2.64	1.326	0.411	14.76	1.226×10^{-3}
5.28	1.326	0.411	14.76	1.239×10^{-3}
7.92	1.326	0.411	14.76	1.414×10^{-3}
15.84	1.326	0.411	14.76	1.554×10^{-3}
Different liquid thermal conductivity of PCM ($\text{W m}^{-1} \text{K}^{-1}$)				
0.58	1.326	0.411	14.76	1.222×10^{-3}
1.16	1.326	0.411	14.76	1.863×10^{-3}
2.32	1.326	0.411	14.76	2.682×10^{-3}
3.48	1.326	0.411	14.76	3.401×10^{-3}
Different initial temperature (K)				
273.15	1.326	0.411	14.76	1.591×10^{-3}
283.15	1.326	0.411	14.76	1.397×10^{-3}
293.15	1.326	0.411	14.76	1.222×10^{-3}
303.15	0.811	0.411	14.76	0.359×10^{-3}
Different hydrogen absorption pressure (MPa)				
0.6	1.326	0.411	14.76	0.840×10^{-3}
0.8	1.326	0.411	14.76	1.222×10^{-3}
1.0	1.326	0.411	14.76	1.431×10^{-3}
1.2	1.326	0.411	14.76	1.700×10^{-3}

3.2 Hydrogen desorption process

3.2.1 Performance comparison of reactors with and without fins

By controlling the same volume of filled metal hydride and phase change material, a metal hydride reactor with hexagonal honeycomb fin and a metal hydride reactor without fin are compared at 0.1MPa and 313.15K. As shown in Figure 13, the metal hydride temperature, PCM temperature, reaction fraction and PCM liquid fraction of the two are compared. It can be observed that the reactor equipped with honeycomb fin releases hydrogen at a significantly faster rate. The metal hydride reactor with hexagonal honeycomb fin has released 90% H_2 in 4560s, compared with the reactor without fin that only released 70% H_2 in 8000s. And the reaction time for the reactor without fins to reach 90% is 6720s. At the initial stage of the reaction, the bed temperature of the two reactors has a similar downward trend. With the reaction going

on, the proposed hexagonal honeycomb fin reactor shows a faster hydrogen release rate. As the reaction progresses, heat transfer becomes more important. Due to the use of fins, the heat required for hydrogen release is absorbed in a short period of time, while the reactor without fin absorbs less heat within a certain period of time. Therefore, as shown in Figure 13a, the temperature rise rate of the reactor without fin is very slow. After reaching the phase transition temperature, the temperature of the hexagonal honeycomb fin reactor gradually increases and tends to equilibrium. Due to the slow heat transfer and temperature rise rate of the reactor without fin, the temperature is only 295K at 8000s, resulting in a slower phase transition rate of PCM. As shown in Figure 13c, the liquid fraction of the reactor without fin is 22%, while that of the hexagonal honeycomb fin reactor is 5%. The final amount of hydrogen gas not released in the reactor without fin is 0.403 wt.%, while the hexagonal honeycomb fin reactor basically releases hydrogen completely, so the developed MH reactor has a faster reaction rate. The comparison of specific KPIs is shown in Table 7. The changes in temperature and reaction fraction at several points in time in the reactor are shown in Figure C.1 of Appendix C.

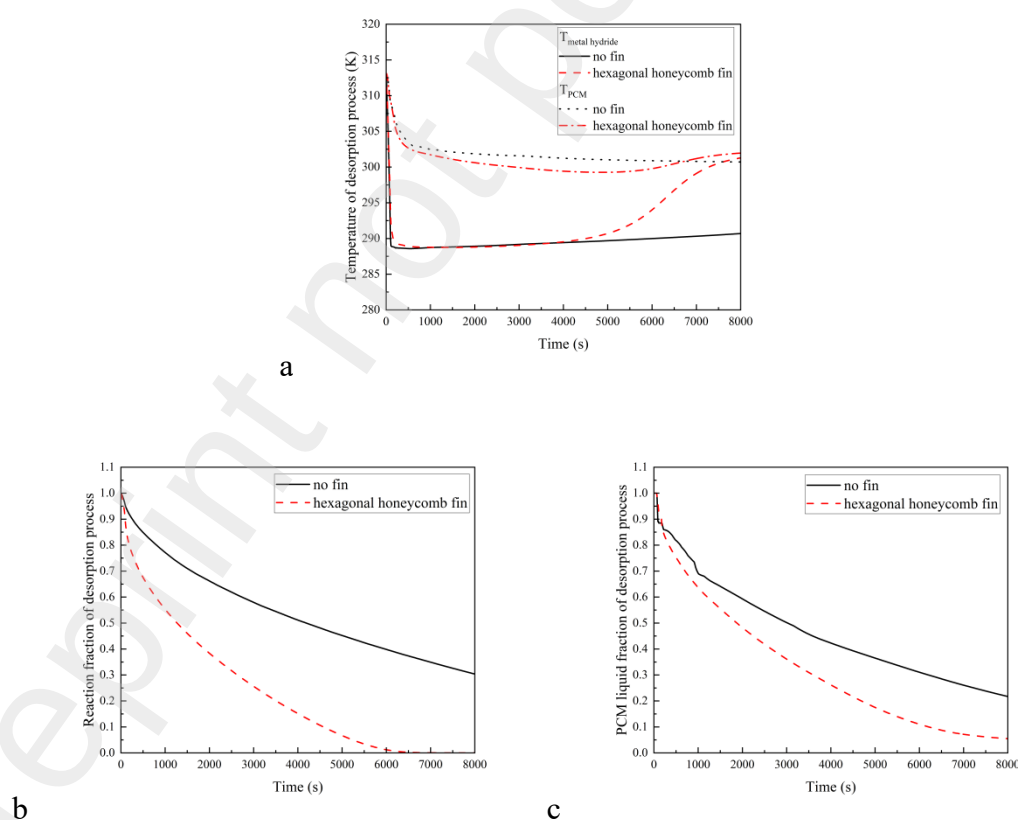


Figure 13 Comparison of (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction in the reactors with and without fins

3.2.2 Effect of fin size change on hydrogen release

The size and form of the reactor are the same as those of the hydrogen absorption process. The reaction fraction, temperature, and liquid phase fraction of three different fin forms are compared at 0.1MPa and initial temperature of 313.15K, as shown in Figure 14. It can be seen that the 6.5mm/0.86mm fin size reaction takes less time, and the hydrogen release rate and temperature reach equilibrium quickly. There is basically no obvious difference in the reaction process between 7.5mm and 8.5mm fins. Compared to these two forms, the 6.5mm fin size reaction rate increases by about 12%, all three fin type reactors achieve 1.326wt.% in 8000s, and the mean saturated rates are $0.868 \times 10^{-3} \text{ g s}^{-1}$, $0.773 \times 10^{-3} \text{ g s}^{-1}$, and $0.766 \times 10^{-3} \text{ g s}^{-1}$ for 6.5mm/7.5mm/8.5mm fins, respectively. Besides, the specific change processes at several points in time in the reactor are shown in Figure C.2 of Appendix C.

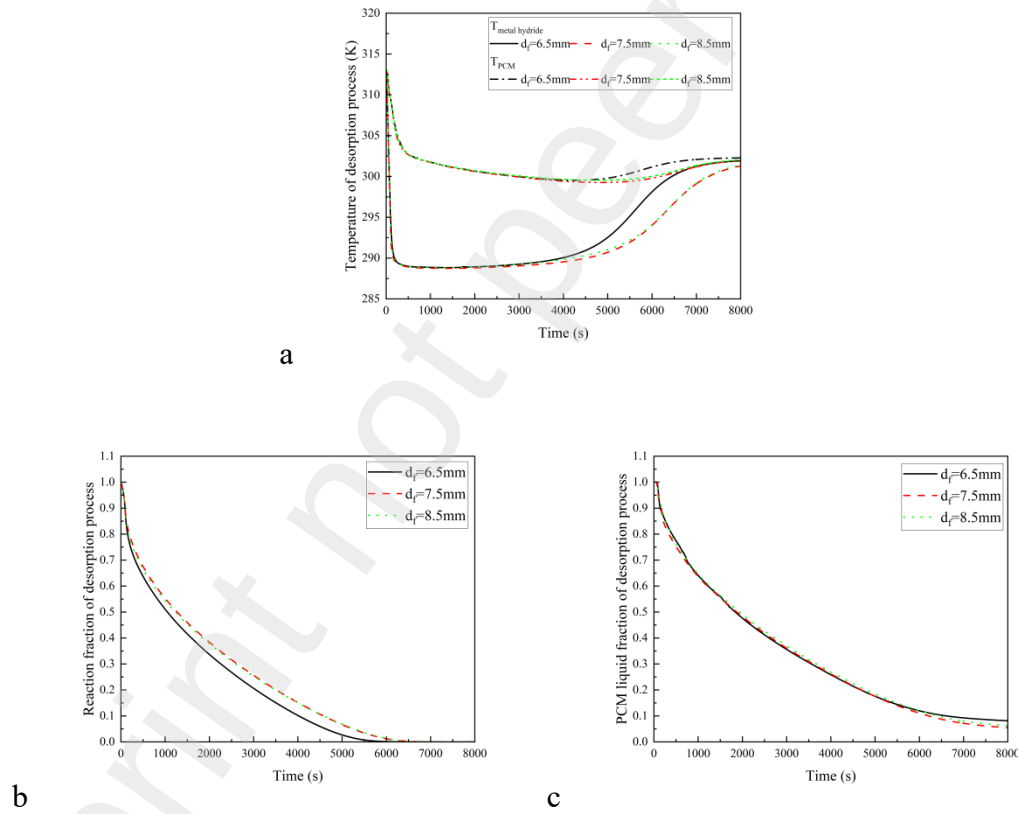


Figure 14 Under different fin sizes, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.2.3 Effect of changes in PCM thickness on hydrogen release

To verify the effect of PCM thickness on the reactor, four reactor forms with the same external height ($h=70\text{mm}$) but different PCM thicknesses ($d_p=11.25\text{mm}$, 11.75mm , 12.25mm , 12.75mm) are compared, and LaNi_5 alloy of the same quality is

used. The changes in hydrogen desorption capacity, PCM liquid fraction, and the temperature of the hydride bed and PCM over time are shown in Figure 15. From the figure, it can be seen that when the thickness of PCM increases from $d_p=11.25\text{mm}$ to $d_p=12.25\text{mm}$, the change in reaction rate is minimal, and the comparison of specific values is shown in Table 7. For all four cases, the hydrogen release amount reached 1.326wt.% in the similar time. However, from the comparison of liquid phase fractions in Figure 15c, the curing ratios of PCM are 97%, 95%, 93%, and 89%, respectively, which shows that the quality of the phase change materials used can make the hydrogen basically complete desorption.

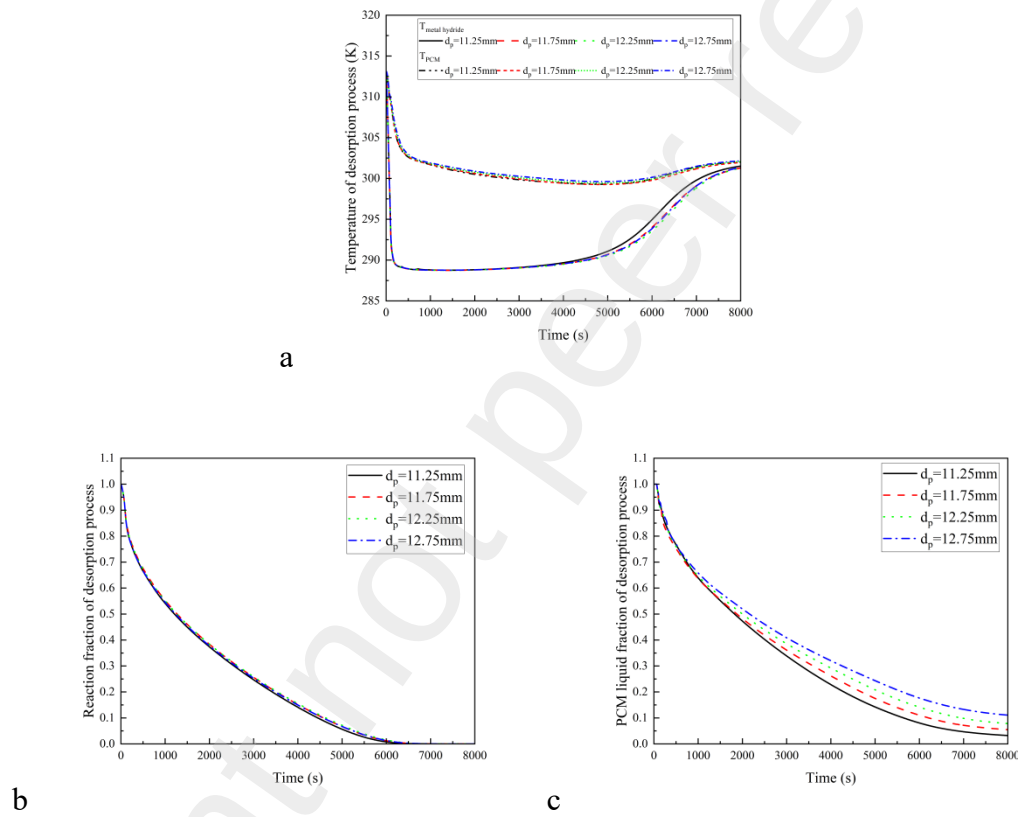


Figure 15 Under different PCM thickness, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.2.4 Effect of changes in operating parameters on hydrogen release

3.2.4.1 Effect of solid thermal conductivity of PCM

By evaluating the influence of different solid phase change materials on the desorption process of reactors, five different thermal conductivities are compared, and the other physical properties remain unchanged. The changes in bed temperature, reaction fraction, phase change material temperature, and liquid phase fraction over time are shown in Figure 16. It is obvious that when the thermal conductivity increases

from $1.32 \text{ W m}^{-1} \text{ K}^{-1}$ to $7.92 \text{ W m}^{-1} \text{ K}^{-1}$, the hydrogen desorption reaction time is significantly reduced. When the thermal conductivity increases to $15.84 \text{ W m}^{-1} \text{ K}^{-1}$, the reaction time is slightly shortened. The time for the reactor to complete 90% of the reaction rate is 4560s, 4170s, 4060s, 3760s, and 3660s, respectively, and the hydrogen desorption time is reduced by about 20%. The corresponding mean saturated rates are $0.773 \times 10^{-3} \text{ g s}^{-1}$, $0.835 \times 10^{-3} \text{ g s}^{-1}$, $0.861 \times 10^{-3} \text{ g s}^{-1}$, $0.920 \times 10^{-3} \text{ g s}^{-1}$, and $0.941 \times 10^{-3} \text{ g s}^{-1}$. In addition, the average temperature of MH bed and PCM is also affected by thermal conductivity. It can be seen from Figure 16a that in the initial stage of the reaction, the temperature drop rate of the MH bed is basically the same. When the thermal conductivity increases from $1.32 \text{ W m}^{-1} \text{ K}^{-1}$ to $15.84 \text{ W m}^{-1} \text{ K}^{-1}$, the temperature reaches the thermal equilibrium state faster. The increase in solid thermal conductivity enables the faster transfer of heat stored in phase change materials to the MH bed, promoting the hydrogen desorption process.

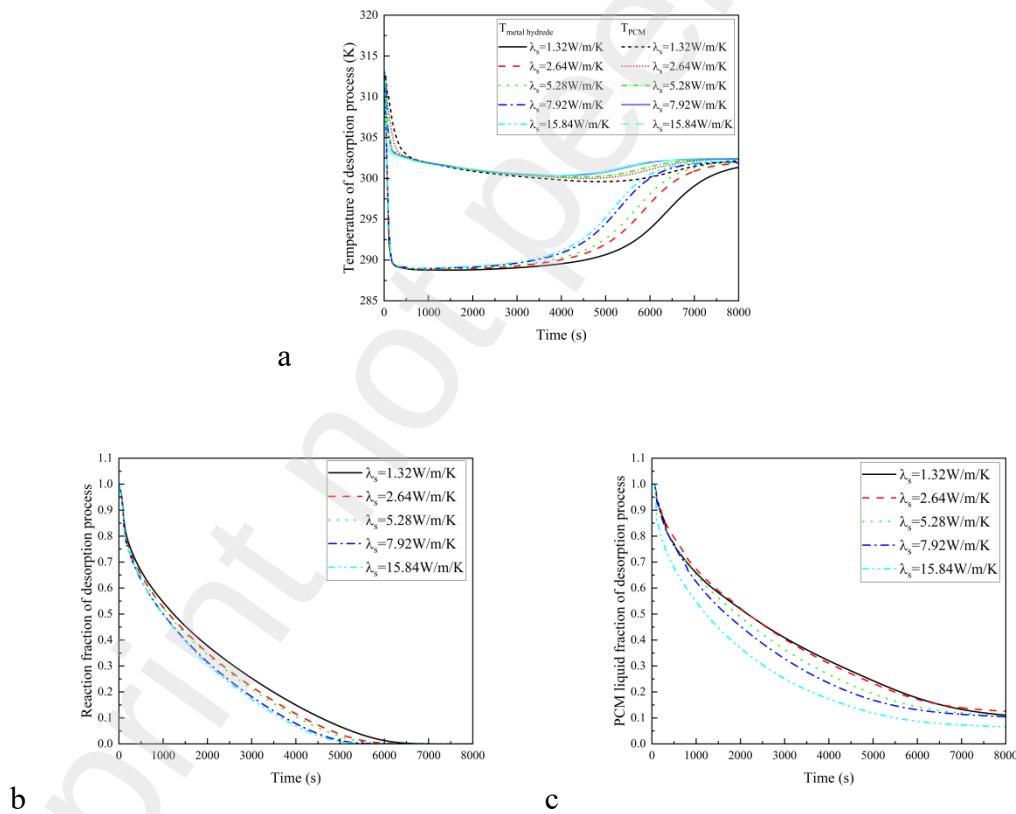


Figure 16 Under different solid thermal conductivity of PCM, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.2.4.2 Effect of liquid thermal conductivity of PCM

When the liquid thermal conductivity is $0.58 \text{ W m}^{-1} \text{ K}^{-1}$, $1.16 \text{ W m}^{-1} \text{ K}^{-1}$, $2.32 \text{ W m}^{-1} \text{ K}^{-1}$ and $3.48 \text{ W m}^{-1} \text{ K}^{-1}$ respectively, the bed temperature, reaction fraction, phase

change material temperature and liquid fraction in the metal hydride reactor are shown in Figure 17. As the liquid thermal conductivity increases, the time required for the bed temperature to decrease to the equilibrium temperature decreases by at least 5370s, and the minimum temperature reached by the bed temperature decreases. The time required for the reactor to complete 90% of the reaction rate is 4560s, 2660s, 1670s, and 1370s, respectively. The hydrogen desorption time decreases by about 70%. The changes of other KPIs are shown in Table 7. It can be seen that the mean saturated rate gradually increases. Similar to the variation pattern of hydrogen absorption process, when the thermal conductivity increases from $2.32\text{W m}^{-1}\text{K}^{-1}$ to $3.48\text{W m}^{-1}\text{K}^{-1}$, the impact on the reaction is limited, the hydrogen desorption time decreases by about 18%.

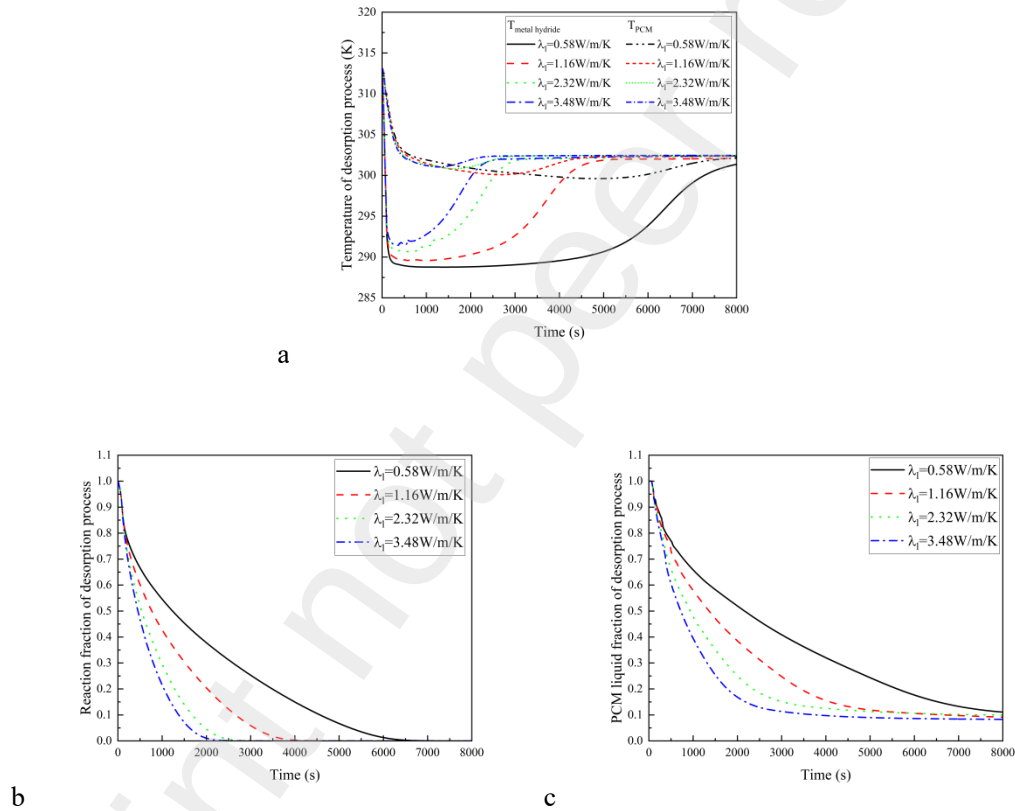


Figure 17 Under different liquid thermal conductivity of PCM, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.2.4.3 Effect of initial temperature

Keeping the other conditions unchanged, the initial temperature values are 293.15K, 303.15K, 313.15K, and 323.15K, respectively, as shown in Figure 18. From the figure, it can be clearly seen that at the initial temperature of 293.15K, the hydrogen release reaction is basically impossible, and the MH temperature and phase change temperature do not change after decreasing to the lowest value. At 8000s, the reaction

fraction remains at 0.94, which is only 6% of the reaction, and the remaining 1.253wt.% hydrogen is not released. Since the bed temperature has been maintained at 287.8K, which does not reach the temperature at which the phase change material occurs. Therefore, as shown in Figure 18c, The liquid fraction at 293.15K has been consistently 0. As the initial temperature increases, the bed temperature increases, and the equilibrium pressure also increases, increasing the reaction driving force and thus accelerating the hydrogen desorption rate. The time required for the reactors to complete 90% of the reaction rate at 303.15K, 313.15K, and 323.15K is 6490s, 4560s, and 3580s, respectively, and the hydrogen desorption time is reduced by about 45%, the corresponding mean saturated rates are $0.575 \times 10^{-3} \text{ g s}^{-1}$, $0.773 \times 10^{-3} \text{ g s}^{-1}$, and $0.931 \times 10^{-3} \text{ g s}^{-1}$, respectively, show a gradually increasing trend. Therefore, a higher initial temperature can promote the reaction.

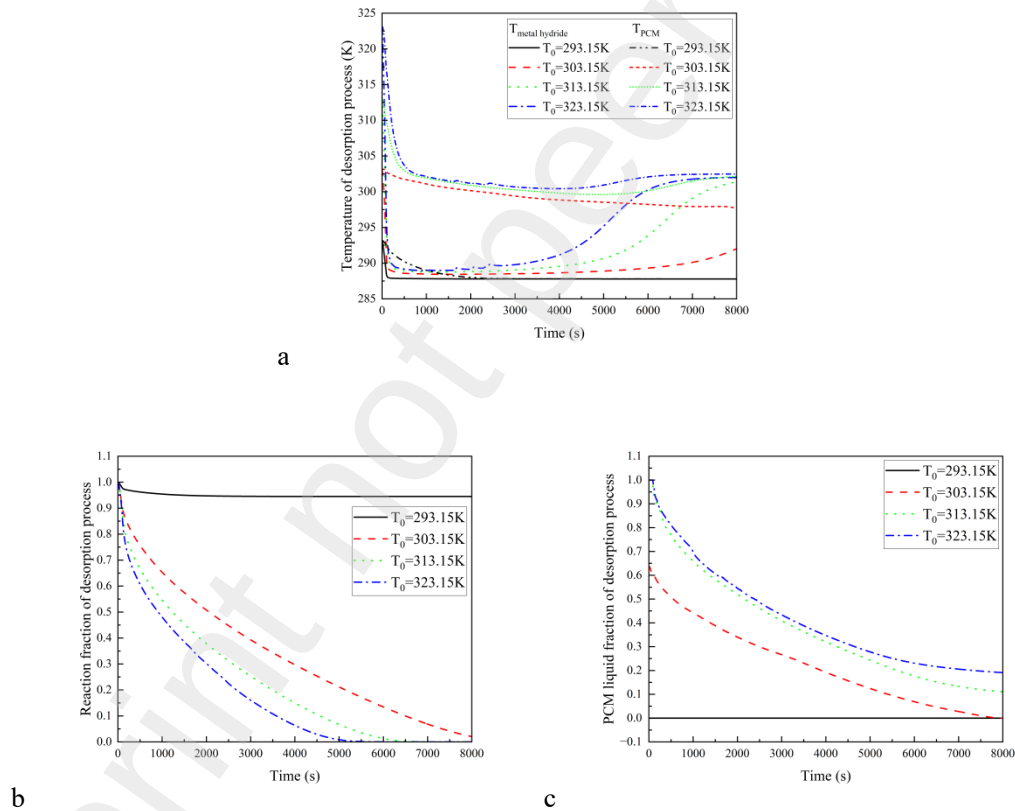


Figure 18 Under different initial temperature, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.2.4.4 Effect of hydrogen desorption pressure

The change of reaction fraction, metal hydride bed temperature, phase change material temperature and PCM liquid fraction with time by changing the hydrogen release pressure is shown in Figure 19. The effect of hydrogen release pressure

(0.025MPa, 0.05MPa, 0.075MPa, 0.1MPa, 0.125MPa) on the hydrogen desorption process is shown. Similar to the absorption process, at the beginning of desorption, the reaction rate is very fast, and the bed temperature drops sharply. The lower the hydrogen release pressure, the more the bed temperature decreases, reaching 265K, 275K, 283K, 288K, and 293K, respectively. This further increases the pressure difference between the hydrogen release pressure and equilibrium pressure during the reaction process, increases the reaction driving force, and promotes the progress of the reaction. The time for the reactor to complete 90% of the reaction rate is 1420s, 2170s, 4290s, 4560s, and 6760s, respectively, the hydrogen desorption time is reduced by about 79%, and the specific mean saturated rate changes are shown in Table 7. Within the studied pressure range, the smaller the hydrogen release pressure, the higher the reaction rate, resulting in an increase in heat provided by PCM and a decrease in its liquid fraction, as shown in Figure 19c.

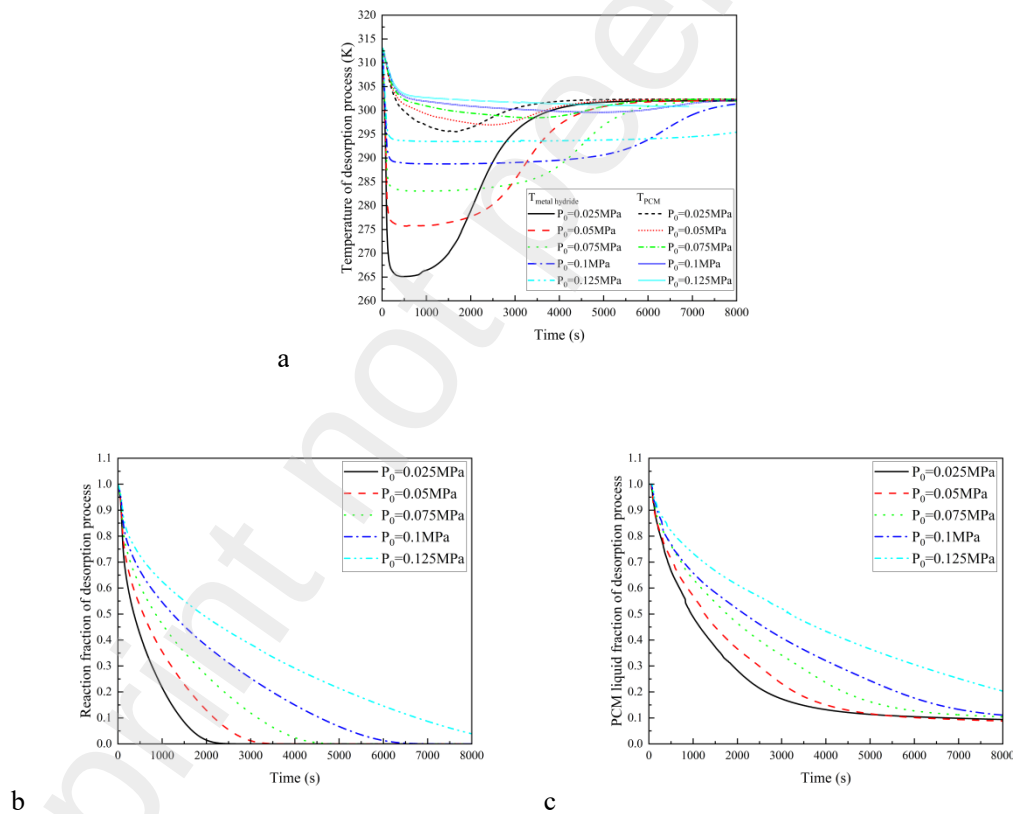


Figure 19 Under different hydrogen absorption pressure, (a) MH temperature and PCM temperature; (b) reaction fraction; (c) liquid fraction of PCM in the reactor change over time

3.2.5 Key performance indicators' results of hydrogen desorption process

The values of weight percentage, system gravimetric storage density, system volumetric storage density and mean saturated rate corresponding to the changes of

different parameters during hydrogen desorption are shown in Table 7.

Table 7 Key performance indicators of hydrogen desorption process

Parameters	wt. %	SGSD (%)	SVSD (kg of H ₂ per m ³)	k (g s ⁻¹)
reactor without fin	0.923	0.642	16.52	0.408×10^{-3}
the developed MH reactor	1.326	0.411	14.76	0.773×10^{-3}
Different fin size (mm)				
6.5	1.326	0.411	14.76	0.868×10^{-3}
7.5	1.326	0.411	14.76	0.773×10^{-3}
8.5	1.326	0.411	14.76	0.766×10^{-3}
Different PCM thickness (mm)				
11.25	1.326	0.417	15.19	0.793×10^{-3}
11.75	1.326	0.411	14.76	0.773×10^{-3}
12.25	1.326	0.405	14.40	0.761×10^{-3}
12.75	1.326	0.399	14.05	0.750×10^{-3}
Different solid thermal conductivity of PCM (W m ⁻¹ K ⁻¹)				
1.32	1.326	0.411	14.76	0.773×10^{-3}
2.64	1.326	0.411	14.76	0.835×10^{-3}
5.28	1.326	0.411	14.76	0.861×10^{-3}
7.92	1.326	0.411	14.76	0.920×10^{-3}
15.84	1.326	0.411	14.76	0.941×10^{-3}
Different liquid thermal conductivity of PCM (W m ⁻¹ K ⁻¹)				
0.58	1.326	0.411	14.76	0.773×10^{-3}
1.16	1.326	0.411	14.76	1.758×10^{-3}
2.32	1.326	0.411	14.76	1.948×10^{-3}
3.48	1.326	0.411	14.76	2.335×10^{-3}
Different initial temperature (K)				
293.15	0.073	0.411	14.76	3.230×10^{-6}
303.15	1.326	0.411	14.76	0.575×10^{-3}
313.15	1.326	0.411	14.76	0.773×10^{-3}
323.15	1.326	0.411	14.76	0.931×10^{-3}
Different hydrogen desorption pressure (MPa)				
0.025	1.326	0.411	14.76	2.257×10^{-3}
0.05	1.326	0.411	14.76	2.153×10^{-3}

0.075	1.326	0.411	14.76	1.481×10^{-3}
0.1	1.326	0.411	14.76	0.773×10^{-3}
0.125	1.326	0.411	14.76	0.564×10^{-3}

4 Conclusion

In this paper, an innovative honeycomb fin metal hydride reactor integrated with PCM is presented, based on LaNi_5 hydrogen storage alloy as medium. A 3D CFD model is established to investigate its performance. In addition, the effects of key factors are also studied. The main conclusions are as follows:

(1) The presence of the fins increases the rate of the hydrogen storage, thanks to the enhanced thermal management of the reactor. Within the reaction time of 8000s, during the absorption process the hydrogen storage capacity of the reactor without fins and with hexagonal honeycomb fins is 1.232wt.% and 1.326wt.%, respectively, and during the desorption process the reactor without fins and with hexagonal honeycomb fins are 0.923wt.% and 1.326wt.%, respectively. The time taken to complete a cycle of hydrogen absorption and desorption is reduced by 12250s. However, due to the presence of the fins, the gravimetric and volumetric storage densities of the proposed MH reactor are lower than those of the reactor without fins (0.411% and 14.76 kg of H_2 per m^3 , against 0.642% and 16.52 kg of H_2 per m^3 , respectively).

(2) During the absorption and desorption processes, the mean saturated rate for the developed reactor is faster than that for the reactor without fins.

(3) In general, the smaller the fin size, the denser the distribution of fins in the reactor, which slightly accelerates the reaction rate. The hydrogen absorption process increases by about 7% and the hydrogen desorption process increases by about 12%.

(4) With the increase of PCM thickness, the total mass and volume of the reactor increase, resulting in a decrease in the gravimetric and volumetric storage densities, and an increase in the mean saturated rates.

(5) The higher the solid and liquid thermal conductivities are, the faster the hydrogen absorption and desorption rates of the reactor, the time for the hydrogen absorption process to reach 90% reaction fraction is reduced by 20% and 71%, respectively, and the hydrogen desorption process is reduced by 20% and 70%.

(6) The lower the initial temperature is, the higher the hydrogen pressure is, the greater the reaction driving force is, the faster the hydrogen absorption rate is, while the slower the hydrogen desorption rate is.

Appendix A. Thermophysical properties of H_2 , LaNi_5 and LiNO_3 -

3H₂O

Table A.1 Thermophysical properties of H₂ and LaNi₅ [13,27,28]

Parameters	Value
Hydrogen absorption reaction rate, C_a (s ⁻¹)	59.187
Hydrogen desorption reaction rate, C_d (s ⁻¹)	9.57
Hydrogen absorption activation energy, E_a (J mol ⁻¹)	21179.6
Hydrogen desorption activation energy, E_d (J mol ⁻¹)	16473
Reaction enthalpy, ΔH (J mol ⁻¹)	30478
Initial density of metal hydride, ρ_{emp} (kg m ⁻³)	7164
Saturated density of metal hydride, ρ_{sat} (kg m ⁻³)	7259
Specific heat of metal hydride, $C_{p,s}$ (J kg ⁻¹ K ⁻¹)	419
Thermal conductivity of metal hydride, λ_s (W m ⁻¹ K ⁻¹)	2.0
Porosity, ε	0.5
Permeability, K (m ²)	10 ⁻⁸
Specific heat of hydrogen, $C_{p,g}$ (J kg ⁻¹ K ⁻¹)	14890
Molar mass of hydrogen, M_g (kg kmol ⁻¹)	2.016
Thermal conductivity of hydrogen, λ_g (W m ⁻¹ K ⁻¹)	0.1815
Universal gas constant, R (J mol ⁻¹ K ⁻¹)	8.314

Table A.2 Thermophysical properties of LiNO₃-3H₂O [15,27,29]

Parameters	Value
Melting temperature, T_{mel} (K)	303
Transition temperature range, ΔT (K)	2
Latent heat, L (kJ kg ⁻¹)	296
Density of liquid phase change material, $\rho_{p,l}$ (kg m ⁻³)	1780
Density of solid phase change material, $\rho_{p,s}$ (kg m ⁻³)	2140
Specific heat of liquid phase change material, $C_{p,p,l}$ (J kg ⁻¹ K ⁻¹)	2770
Specific heat of solid phase change material, $C_{p,p,s}$ (J kg ⁻¹ K ⁻¹)	1730
Thermal conductivity of liquid phase change material, $\lambda_{p,l}$ (W m ⁻¹ K ⁻¹)	0.58
Thermal conductivity of solid phase change material, $\lambda_{p,s}$ (W m ⁻¹ K ⁻¹)	1.32

Appendix B. Hydrogen absorption process

Parameters	Hydrogen absorption process
	t=200s

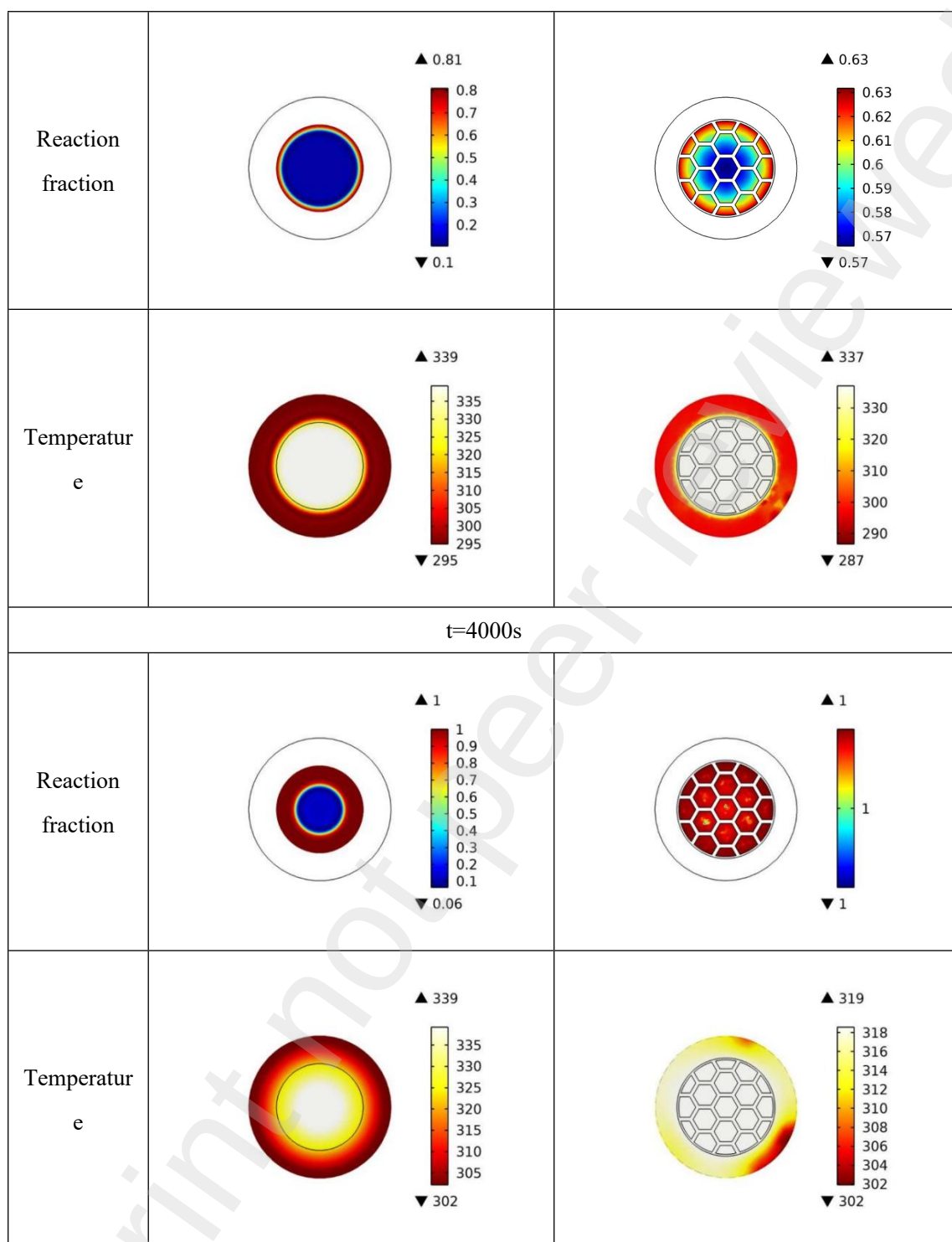
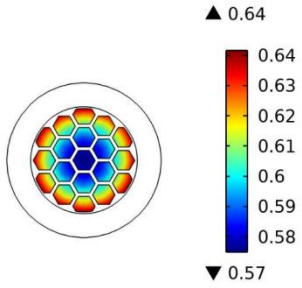
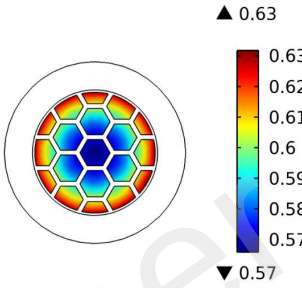
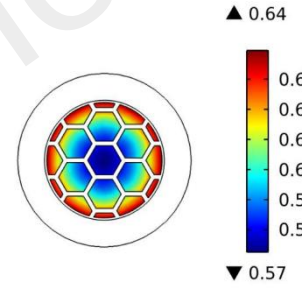
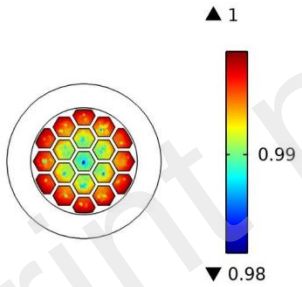
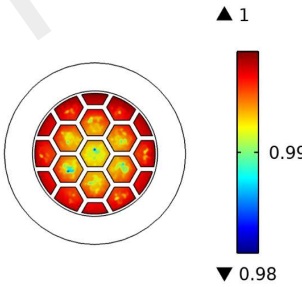
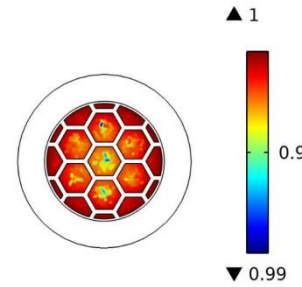
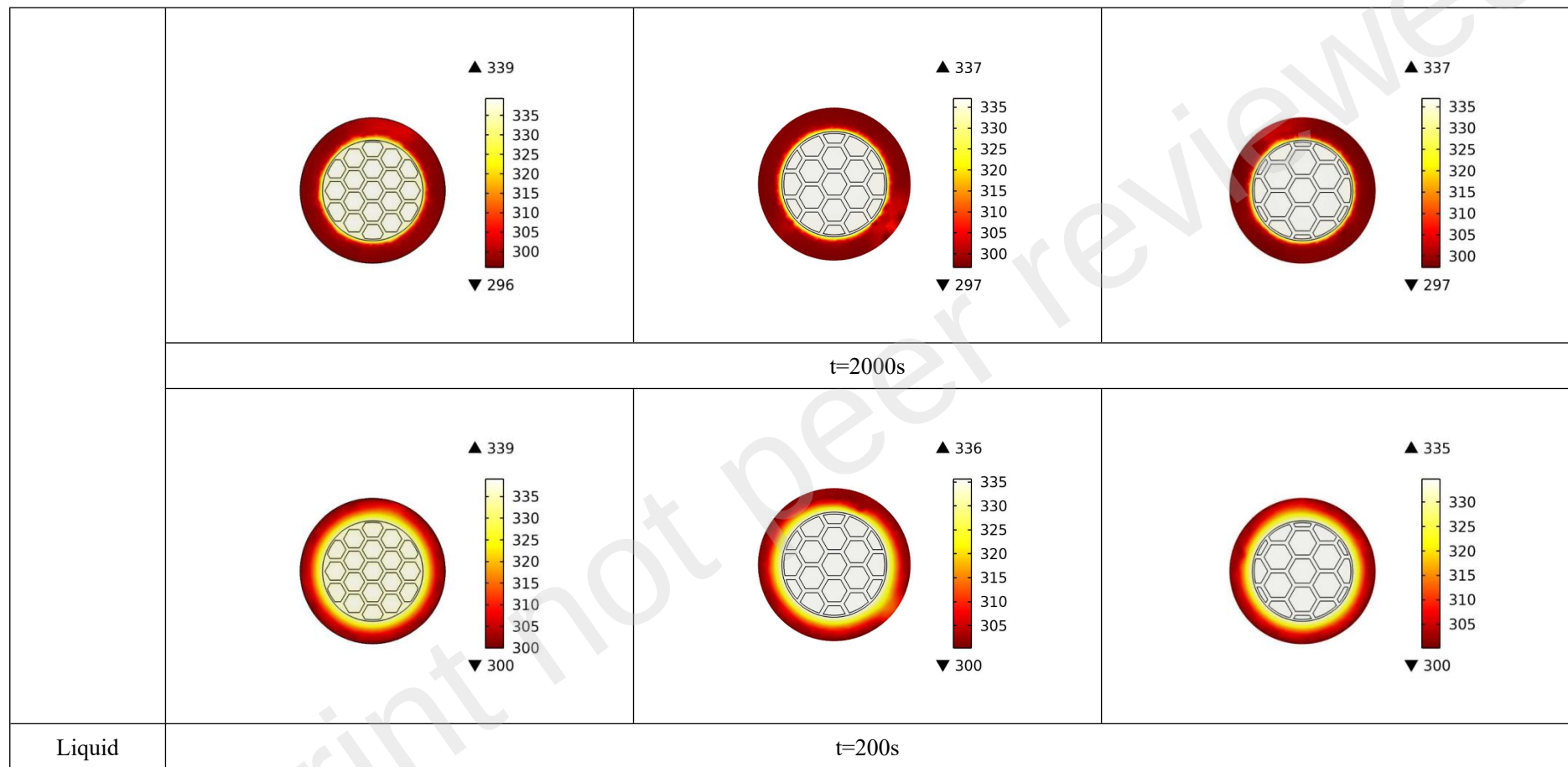


Figure B.1 Distribution of temperature and reaction fraction in the reactors with and without fins in hydrogen absorption process

Parameters	6.5mm	7.5mm	8.5mm
Reaction fraction	t=200s		
			
	t=2000s		
			
Temperature	t=200s		



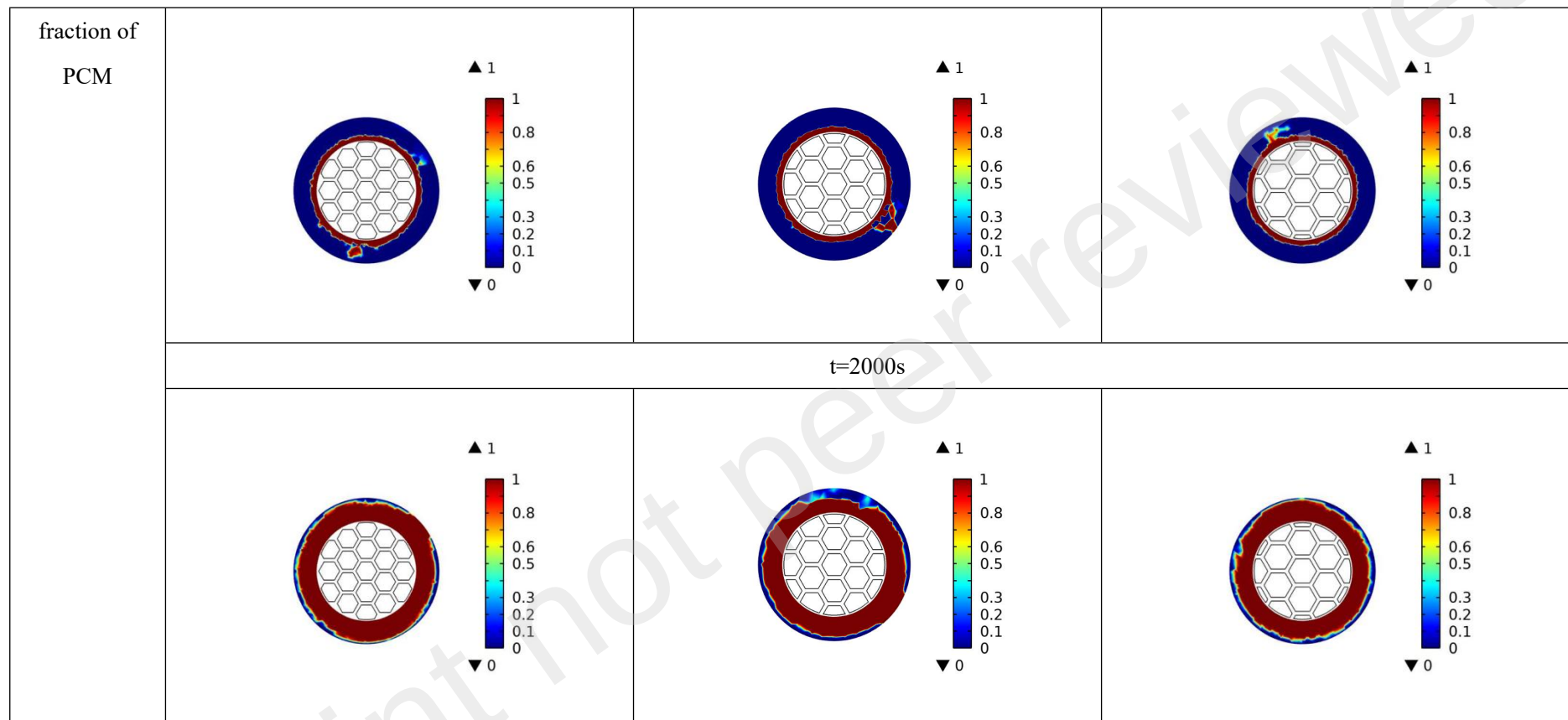
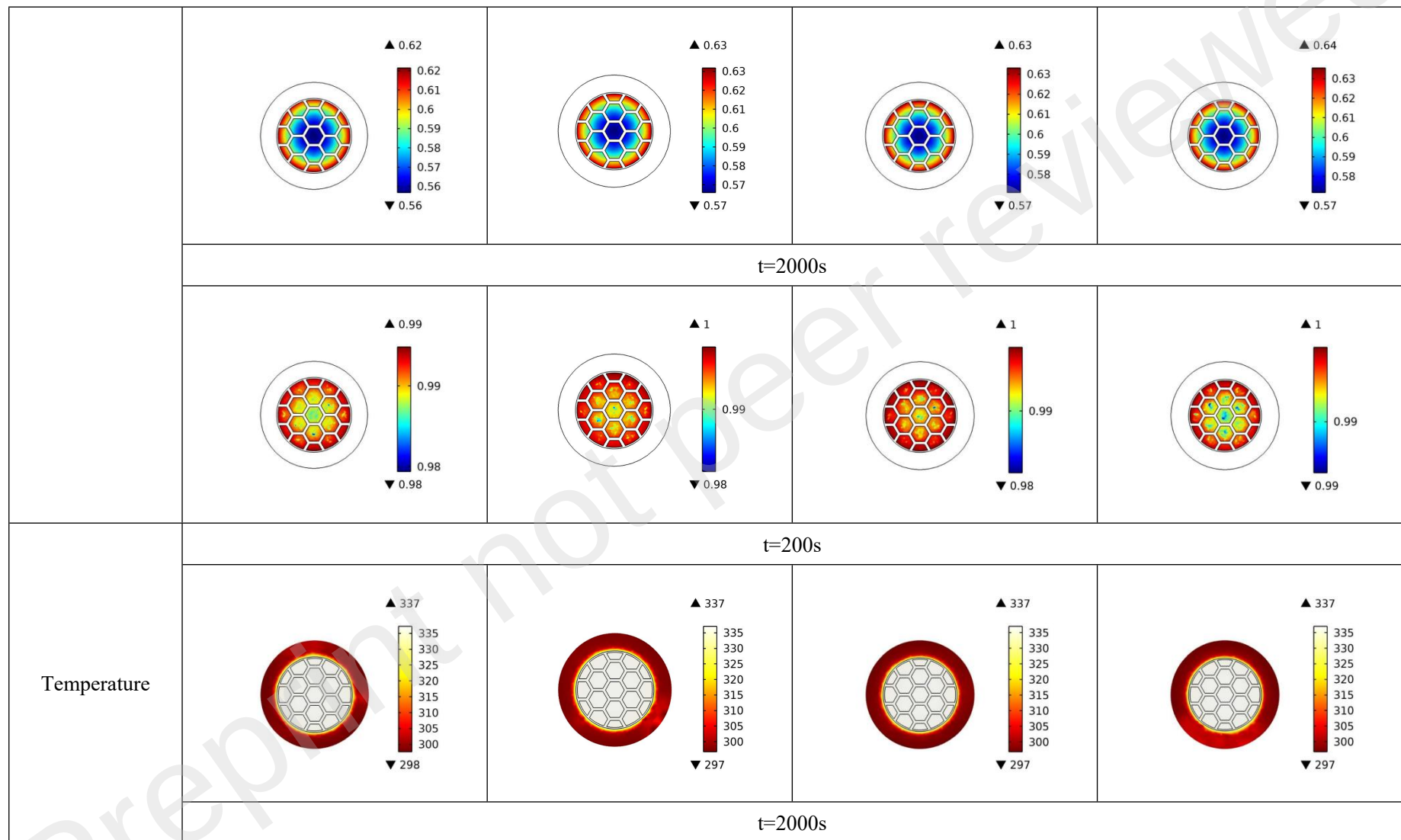


Figure B.2 Under different fin sizes, comparison of the reaction fraction, MH temperature, and liquid fraction of PCM of the reactor at 200s and 2000s

Parameters	11.25mm	11.75mm	12.25mm	12.75mm
Reaction fraction	t=200s			



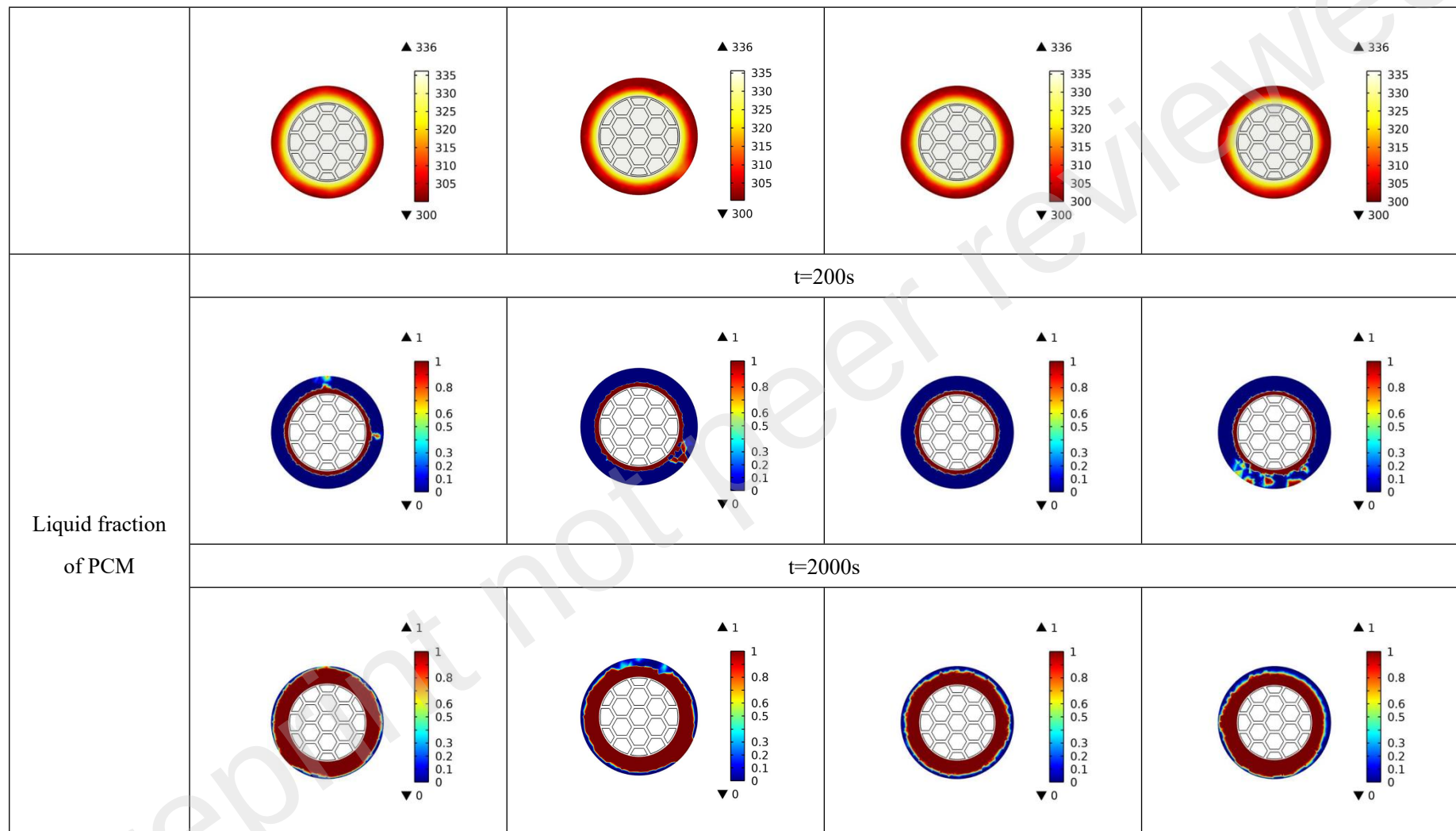


Figure B.3 Under different PCM sizes, comparison of reaction fraction, MH temperature, and liquid fraction of PCM of the reactors at 200s and 2000s

Appendix C. Hydrogen desorption process

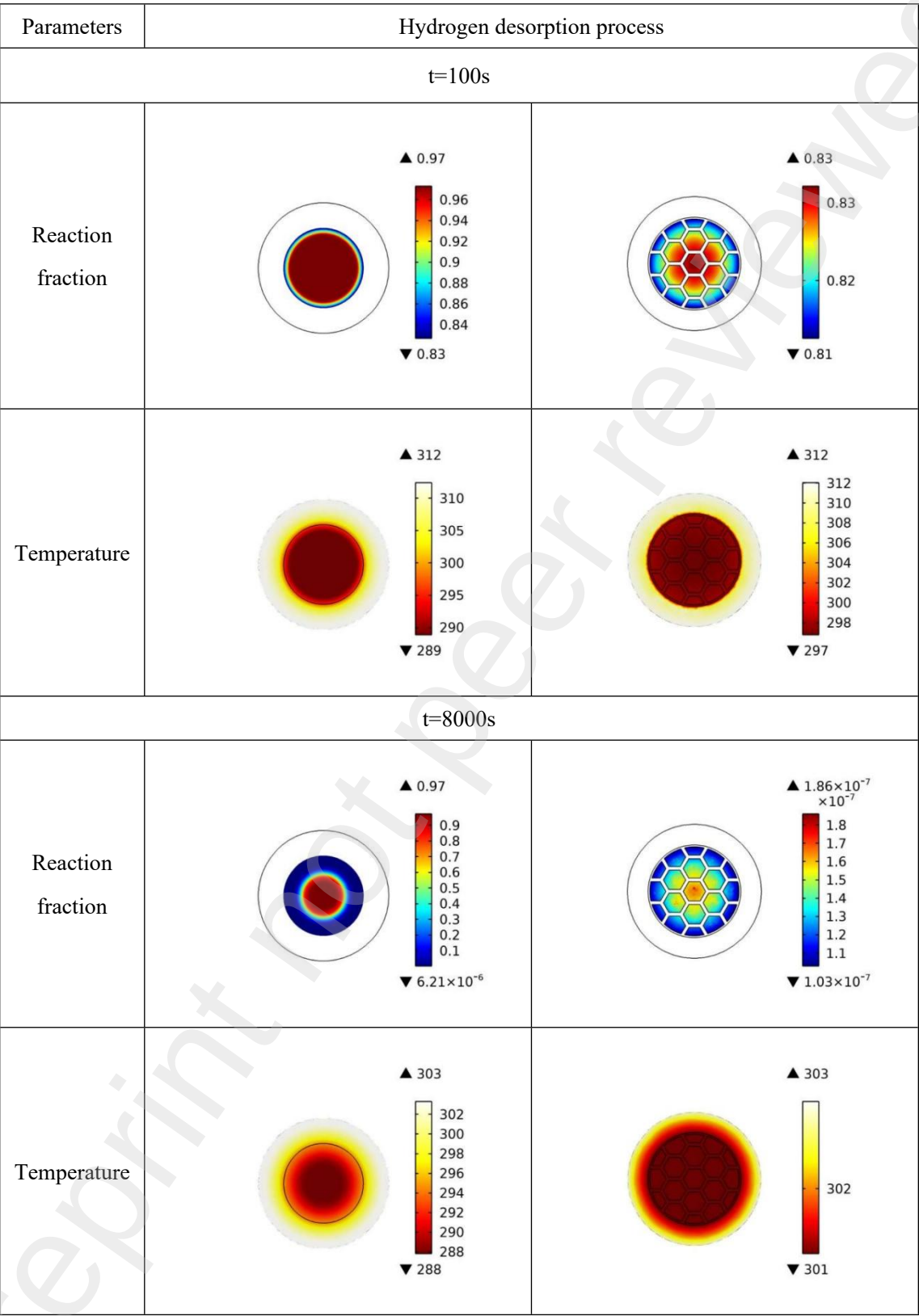
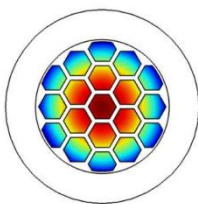
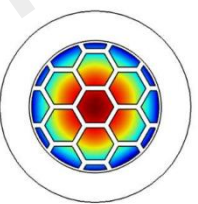
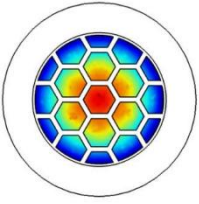
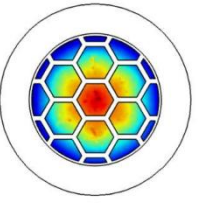
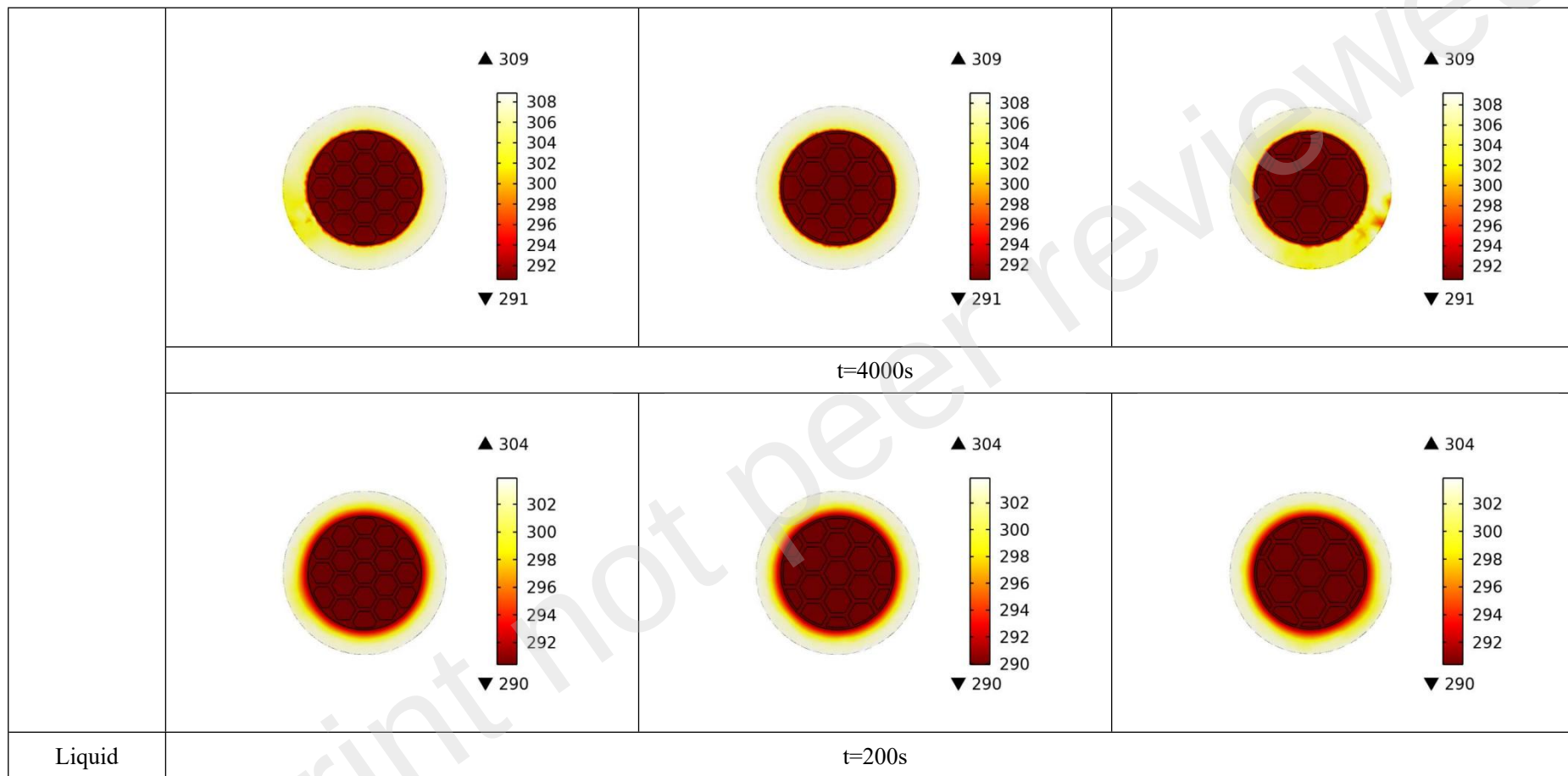


Figure C.1 Distribution of temperature and reaction fraction in the reactors with and without fins in hydrogen desorption process

Parameters	6.5mm	7.5mm	8.5mm
Reaction fraction	t=200s		
	 ▲ 0.64 ▼ 0.6	 ▲ 0.69 ▼ 0.65	 ▲ 0.67 ▼ 0.62
	t=4000s		
	 ▲ 0.03 $\times 10^{-2}$ ▼ 0.02	 ▲ 0.05 $\times 10^{-2}$ ▼ 0.03	 ▲ 0.03 $\times 10^{-2}$ ▼ 0.02
Temperature	t=200s		



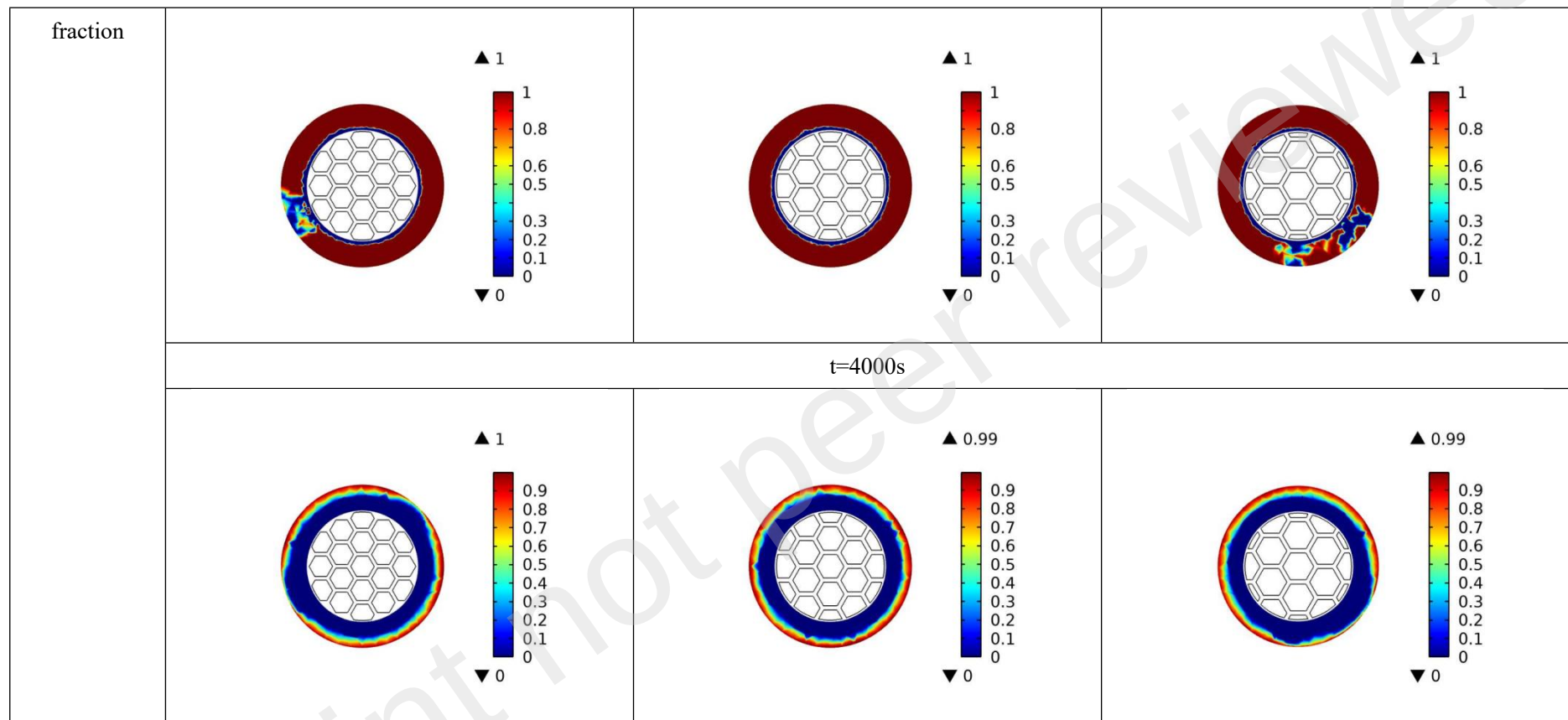


Figure C.2 Under different fin sizes, comparison of the reaction fraction, MH temperature, and liquid fraction of PCM of the reactor at 200s and 4000s

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